

The image shows a well-lit industrial networking lab. In the foreground, there is a long wooden workbench on a black frame with many drawers. A black office chair is positioned in front of it. In the background, there are several rows of equipment racks filled with various electronic components and cables. A projector is mounted on the ceiling, and the room has a clean, professional appearance.

INDUSTRIAL NETWORKING MENG 4010

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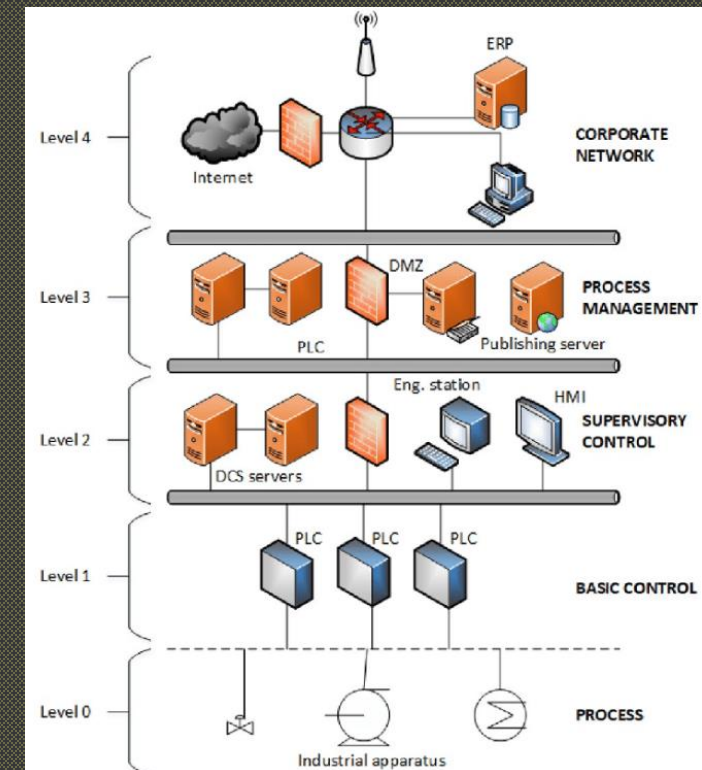
8B.1 Introduction to Industrial Networks

Industrial networking is the backbone of modern automation systems. It allows controllers, field devices, and supervisory systems to exchange data efficiently, reliably, and in real time. Unlike office networks, industrial networks are designed to operate under harsh conditions such as high temperatures, electromagnetic interference, and mechanical vibrations.

- Example: A factory line uses multiple PLCs to control motors, sensors, and actuators. Industrial networking ensures that a motor receives a start command exactly when a sensor detects a product in position.

Key Points:

- Enables centralized and decentralized control
- Supports both monitoring and control functions
- Must ensure deterministic communication for safety and performance



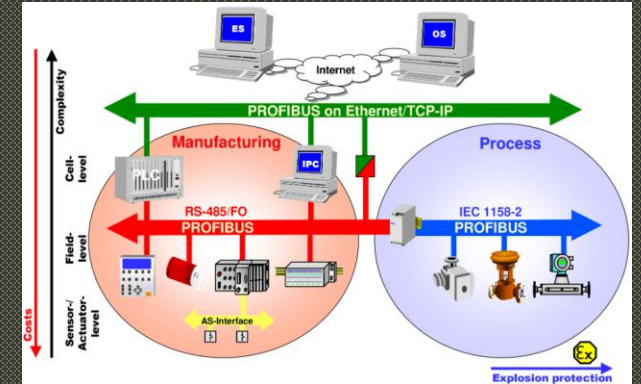
8B.1 Introduction to Industrial Networks

Fieldbus

- Serial communication system connecting controllers to field devices.
- Uses master-slave communication (PROFIBUS) with deterministic timing.
- Suitable for discrete and analog process data.

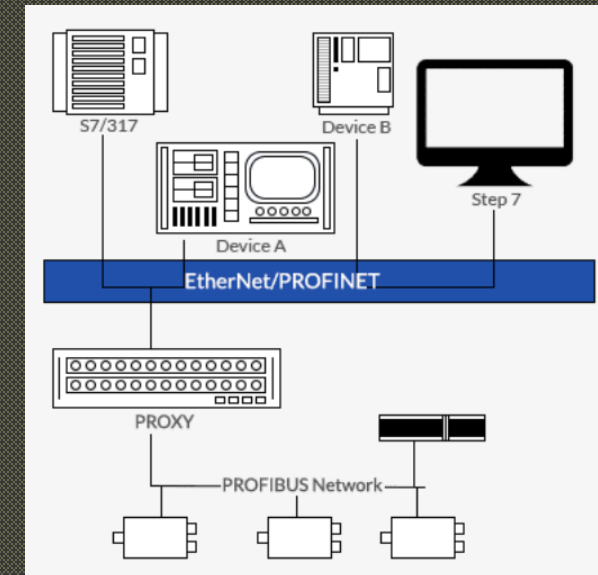
Ethernet-Based Networks

- High-speed communication using standard Ethernet protocols.
- Supports peer-to-peer communication (PROFINET) and real-time control.
- Can integrate with office networks for monitoring and reporting.



<https://instrumentationtools.com/fieldbus-protocols/>

Feature	Fieldbus (PROFIBUS)	Ethernet (PROFINET)
Physical Layer	RS-485, MBP	Twisted pair, fiber optic
Max Devices per Segment	32	Depends on switch topology
Data Rate	Up to 12 Mbps	100 Mbps – 1 Gbps
Communication Type	Master-Slave	Peer-to-Peer / Controller-Device
Determinism	High (cyclic communication)	RT & IRT modes available
Typical Applications	Factory & process automation	High-speed automation, motion control



<https://www.rtautomation.com/technologies/profinet-io/>

8B.1 Introduction to Industrial Networks

Overview of PROFIBUS

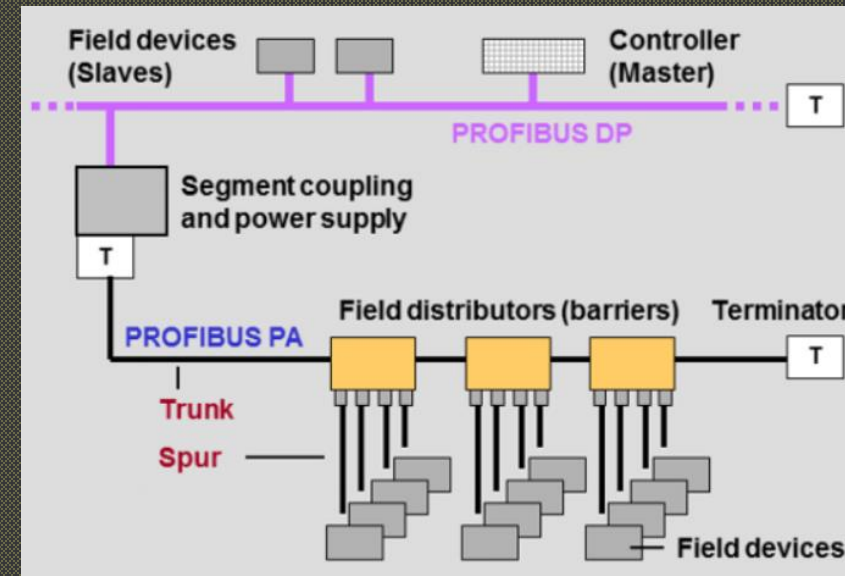
PROFIBUS (Process Field Bus) is a serial communication standard widely used in industrial automation. It connects controllers (masters) with field devices (slaves) for process control and monitoring.

Features:

- Master-slave communication with token passing
- Supports multiple protocol types: DP, PA, PROFIsafe, PROFINET
- Deterministic and robust in electrically noisy environments

Example: A chemical plant uses PROFIBUS PA to connect pressure and temperature transmitters to a DCS. Each transmitter communicates its readings cyclically to ensure safe control.

PROFIBUS FMS (Fieldbus Message Specification) is a communication protocol for complex, non-deterministic data exchange between intelligent devices like PLCs and PCs in industrial automation



8B.1 Introduction to Industrial Networks

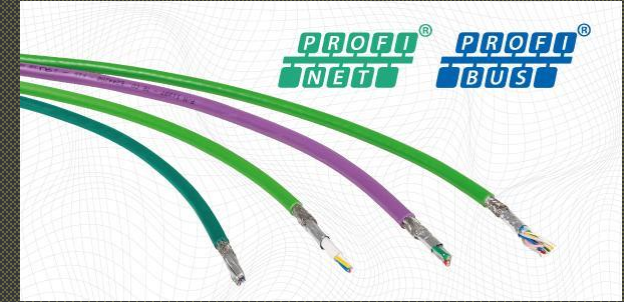
Overview of PROFINET

PROFINET is an Ethernet-based industrial network designed for real-time communication.

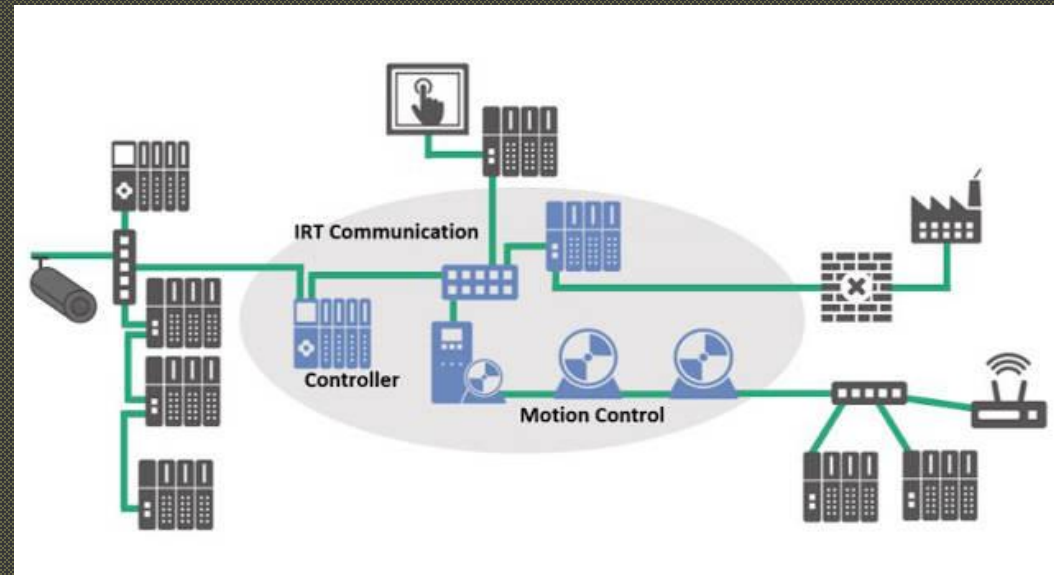
Features:

- Full-duplex Ethernet with RT (Real-Time) and IRT (Isochronous Real-Time) modes
- Supports IO-Controllers, IO-Devices, and IO-Supervisors
- Facilitates peer-to-peer communication and vertical integration
- Compatible with other fieldbuses via proxy gateways

Example: A packaging machine uses PROFINET IRT to synchronize multiple servo motors, ensuring precise product placement at high speed.



<https://control.com/technical-articles/understanding-profibus-vs-profinet/>



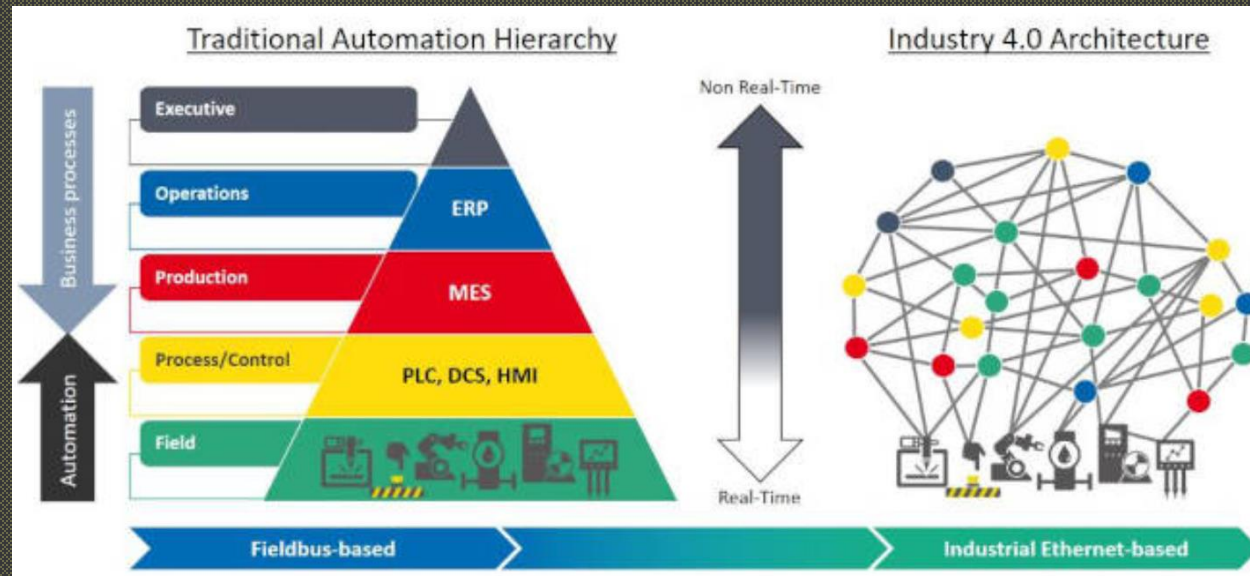
<https://www.controleng.com/understanding-the-difference-between-profinet-rt-and-profinet-irt/>

8B.1 Introduction to Industrial Networks

Industrial networking improves efficiency, reliability, and safety in automation systems:

- **Cost Reduction:** Minimizes wiring by replacing point-to-point connections with a network.
- **Faster Commissioning:** Devices can be configured and diagnosed over the network.
- **Higher Accuracy:** Digital communication avoids conversion errors between analog and digital signals.
- **Asset Management:** Continuous monitoring of field devices enables predictive maintenance.

Example: Using PROFINET, a plant manager can detect that a valve actuator is degrading before it fails, avoiding costly downtime.



8B.2 PROFIBUS: DP, PA, and PROFIsafe

PROFIBUS DP (Decentralized Peripherals) is designed for high-speed communication between controllers (masters) and distributed field devices (slaves) in factory automation.

Key Features:

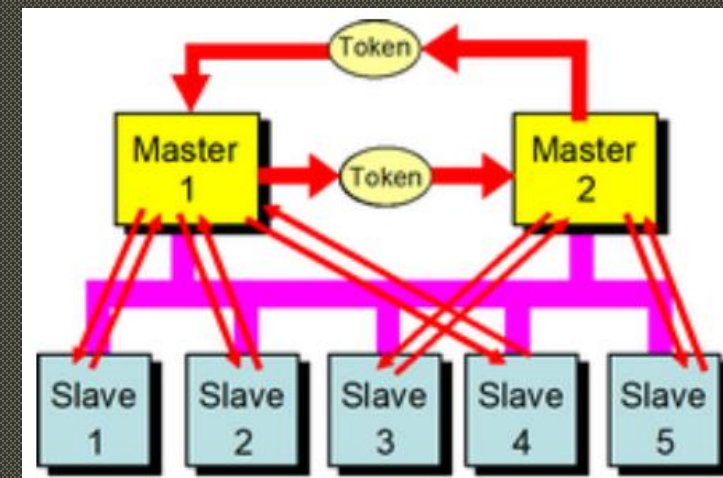
- Uses RS-485 physical layer for deterministic communication
- Supports data rates up to 12 Mbps
- Handles large data packets (up to 244 bytes)
- Deterministic cyclic data exchange

Example: A bottling plant uses PROFIBUS DP to connect a PLC with multiple I/O modules controlling conveyor belts, pumps, and sensors. The PLC receives sensor status every 2 ms for precise coordination.

There are two classes of master devices for PROFIBUS networks, class 1 and class 2.

- A class 1 master is the “main” master device, responsible for cyclically requesting data from every slave.
- A class 2 master acyclically communicates with slave devices for the purposes of diagnostics and setting device parameters.

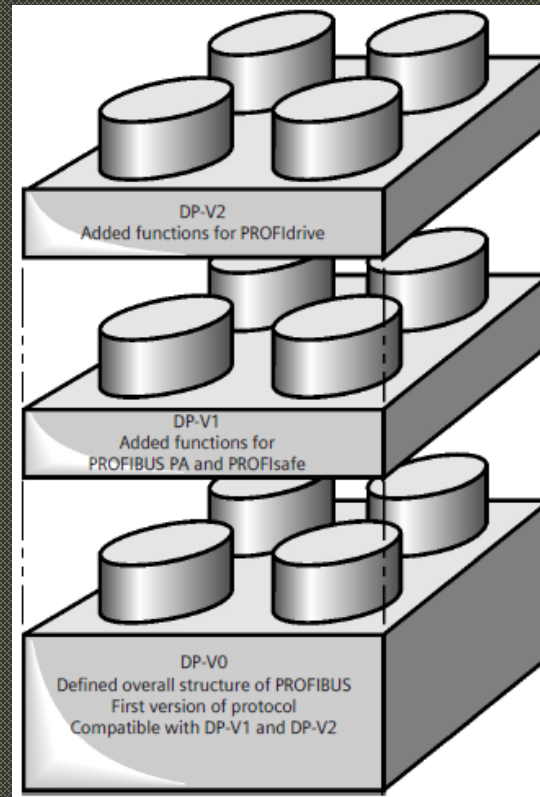
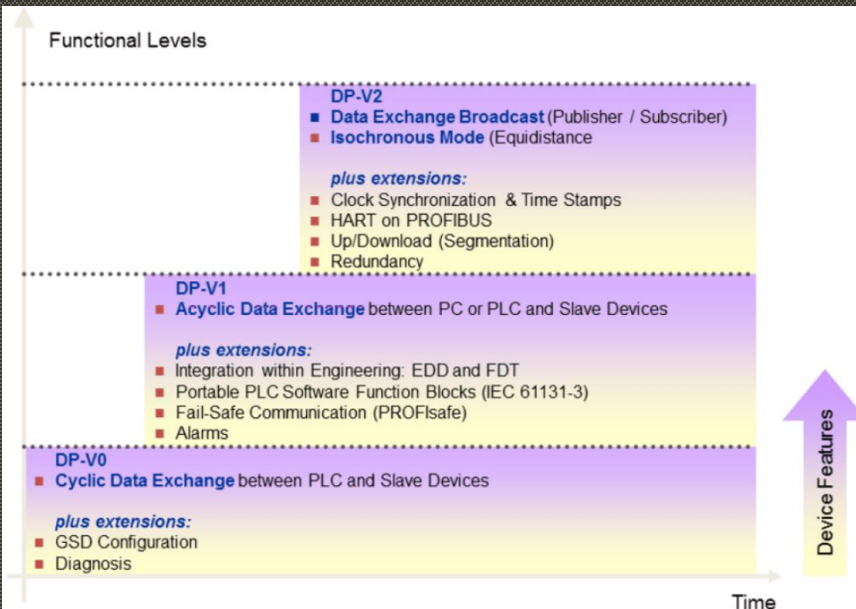
The term “decentralized periphery” is derived from the structure of a PROFIBUS DP system. Instead of having every single field device in a factory connected to the programmable logic controller (PLC) directly via extensive cabling, PROFIBUS DP decentralizes the I/O connections and brings them to the factory floor.



8B.2 PROFIBUS: DP, PA, and PROFIsafe

PROFIBUS DP Protocol Versions

- DP-V0: Basic protocol for reading/writing time-critical data
- DP-V1: Adds support for process automation and safety functions (integration with PA and PROFIsafe)
- DP-V2: Adds high-speed drive control functions while remaining backward-compatible



Version	Key Features	Applications
DP-V0	Basic cyclic data exchange	Standard factory automation
DP-V1	Acyclic communication, safety support	Process automation
DP-V2	High-speed drives, extended functions	Multi-axis motion control

It operates on the physical layer RS-485 - the same physical layer as Modbus - but it has a specialized token ring protocol that enables multiple master devices to exist on the same network, differentiating it from other bus networks.

A “token” is passed between masters, and whichever master is holding the token has the exclusive ability to communicate with slave devices on the network.

8B.2 PROFIBUS: DP, PA, and PROFIsafe

PROFIBUS PA (Process Automation) is tailored for process industries, connecting field instruments such as pressure, temperature, and level transmitters.

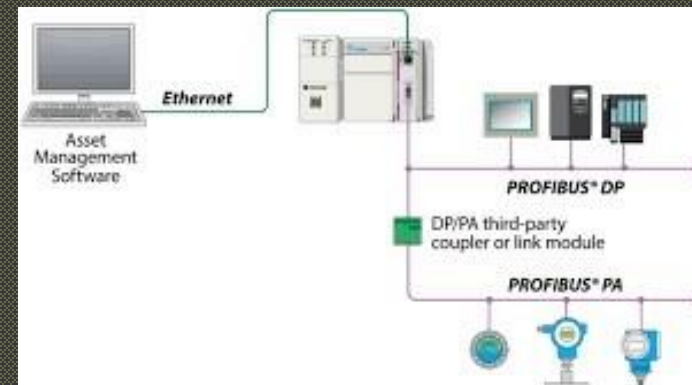
Features:

- Uses Manchester-encoded Bus Powered (MBP) physical layer
- Provides power and communication over the same two-wire cable
- Intrinsically safe for hazardous environments
- Configurable over the bus

Example: In a chemical plant, pressure transmitters powered over MBP communicate readings to a DCS without needing additional power cables, reducing installation costs and simplifying maintenance.

DCS – Distributed Control System

- Manchester-encoded Bus Powered (MBP) is a physical layer standard for the PROFIBUS PA network, which transmits both power and data over a single two-wire cable.
- This bus-powered approach simplifies installation by eliminating the need for separate power supplies at field devices, making it ideal for process industries and hazardous environments.
- It uses Manchester encoding at a fixed baud rate of 31.25 kbit/s.

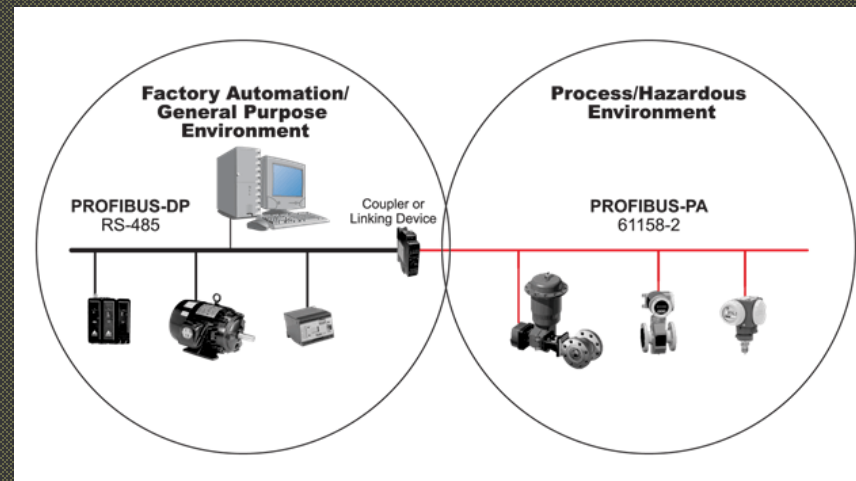


8B.2 PROFIBUS: DP, PA, and PROFIsafe

PROFIBUS PA Features

- Bus Power: Supplies energy and communication via one cable.
- Safe Design: Low voltage/current prevents sparks in explosive environments.
- Device Profiles: Standardized parameters across devices for easy integration.
- Remote Configuration: Devices can be configured and diagnosed over the network.

Feature	DP (RS-485)	PA (MBP)
Power	External	Bus-powered
Max Devices per Segment	32	32 (non-IS) / 24 (IS)
Distance	1200 m	1900 m (non-IS) / 1000 m (IS)
Safety	Not intrinsically safe	Intrinsically safe
Baud Rate	Up to 12 Mbps	31.25 kbps



8B.2 PROFIBUS: DP, PA, and PROFIsafe

PROFIsafe is a safety extension of PROFIBUS and PROFINET, suitable for safety-critical applications up to SIL-3 level (IEC 61508).

Features:

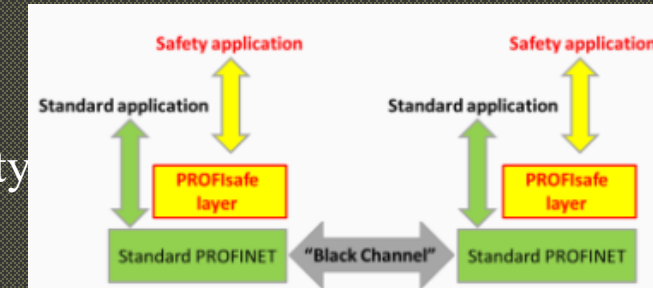
- Detects errors such as message repetition, loss, corruption, or incorrect sequencing
- Uses message numbering and loopback for error detection
- Integrates seamlessly with standard PROFIBUS/PROFINET networks

SIL 3 is the third-highest level of Safety Integrity Level, a measure of the effectiveness of a safety system in reducing the risk of a specific hazard

Example: In a packaging line, PROFIsafe monitors emergency stop circuits. If a cable fault occurs, the system safely halts all machinery without risk of maloperation.

Safety Functions

- Error Detection: Loopback, consecutive numbering, and timeouts ensure message integrity
- Device Verification: Confirms correct transmitter-receiver pairings.
- Watchdog Monitoring: Stops operations if communication fails within defined intervals.



<https://profinetuniversity.com/functional-safety/profisafe-profile-overview/>

Mechanism	Function
Loopback	Confirms message received correctly
Consecutive Numbering	Tracks order of messages to prevent sequence errors
Time Expectation	Ensures messages arrive within expected time
Device Identification	Validates correct device communication

- Loopback Test: A PLC sends a signal to a remote I/O module, which immediately returns the signal back. If the returned data doesn't match the original, the PLC detects a transmission error.
- Consecutive Numbering: Each safety message includes a sequence number. If a number is missing or out of order, the receiver detects a lost or duplicated message.
- Timeouts: A maximum allowed time is set for receiving a response. If the response does not arrive within that time, a communication failure is declared.

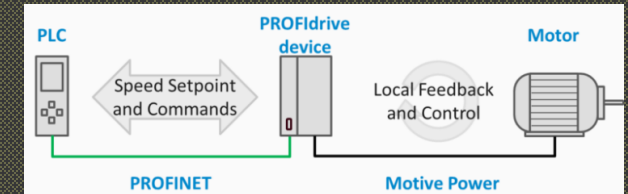
8B.2 PROFIBUS: DP, PA, and PROFI-safe

PROFIdrive extends PROFIBUS and PROFINET for high-speed drive applications.

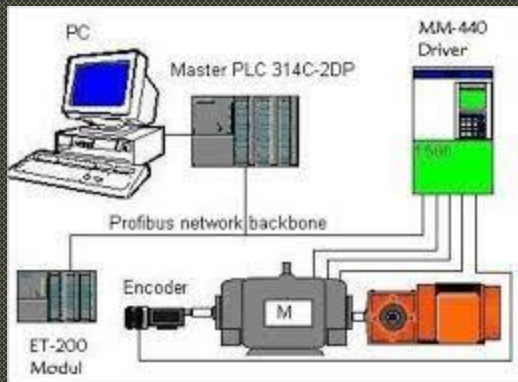
Key Features:

- Equidistant cycle time for precise control
- Clock synchronization across multiple axes
- Supports slave-to-slave communication (publisher/subscriber model)
- Backward-compatible with standard PROFIBUS DP

Example: A CNC machine with multiple servo motors uses PROFIdrive to synchronize spindle and axis movements, achieving precise machining with minimal jitter.



<https://profinetuniversity.com/application-profiles/profidrive-application-class-1/>



Protocol	Physical Layer	Typical Applications	Key Features
DP	RS-485	Factory automation	High-speed, cyclic communication
PA	MBP	Process automation	Bus-powered, intrinsically safe
PROFI-safe	DP or PA	Safety-critical systems	SIL-3, error detection, safe messaging
PROFIdrive	DP, Ethernet (PROFINET)	High-speed drives	Equidistant cycle, multi-axis sync

https://www.researchgate.net/figure/The-block-scheme-of-the-profibus-based-AM-control_fig1_319893823

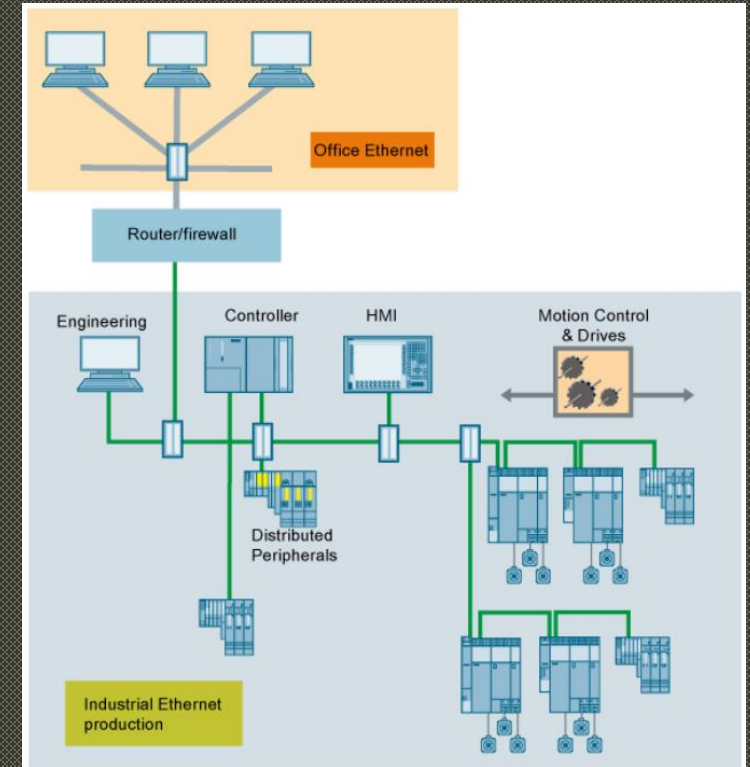
8B.3 PROFINET Basics, Device Types, and Communication Models

PROFINET is an open industrial Ethernet standard designed for high-speed, real-time communication in automation systems. It integrates controllers, field devices, and engineering tools over a single network.

Key Features:

- Uses standard Ethernet cabling (twisted pair or fiber)
- Supports full-duplex communication for simultaneous data transfer
- Real-time (RT) and Isochronous Real-Time (IRT) communication modes
- Supports peer-to-peer, vertical, and horizontal integration
- Conforms to open standards, allowing cross-vendor interoperability

Example: In an automotive assembly plant, PROFINET connects multiple PLCs, robot controllers, drives, and sensors to coordinate high-speed production lines.



https://www.researchgate.net/figure/The-block-scheme-of-the-profinet-based-AM-control_fig1_319893823

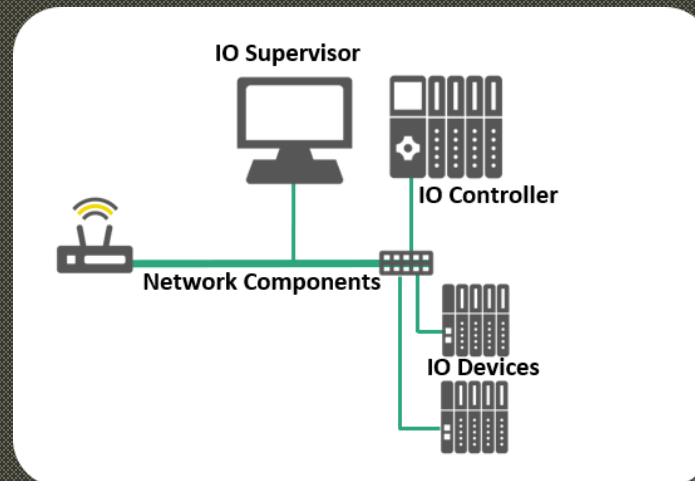
8B.3 PROFINET Basics, Device Types, and Communication Models

PROFINET defines three main device roles:

Device Type	Function	PROFIBUS Equivalent
IO-Controller	Controls process and manages I/O communication	Class 1 Master
IO-Device	Field device providing input/output data	Slave
IO-Supervisor	Engineering station for configuration/diagnostics	Class 2 Master

Example: A conveyor system uses:

- IO-Controller: PLC managing all motor drives
- IO-Devices: Sensors and actuators on the conveyor
- IO-Supervisor: HMI/engineering station for monitoring and maintenance



8B.3 PROFINET Basics, Device Types, and Communication Models

Communication Models

1. Consumer/Provider Model

- IO-Device provides input data (e.g., sensor readings)
- IO-Controller consumes input data
- IO-Controller provides output data (e.g., motor commands)
- IO-Device consumes output data

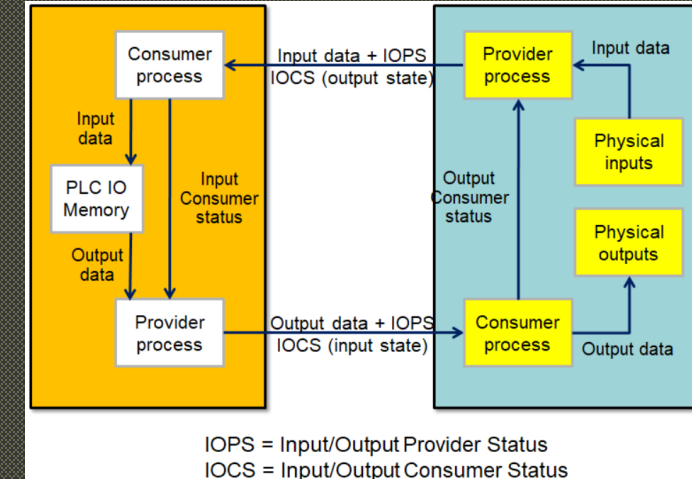
2. Cyclic Communication

- Time-sensitive I/O data exchanged continuously
- Managed via scheduled update cycles
- Priority given to real-time messages over standard Ethernet traffic

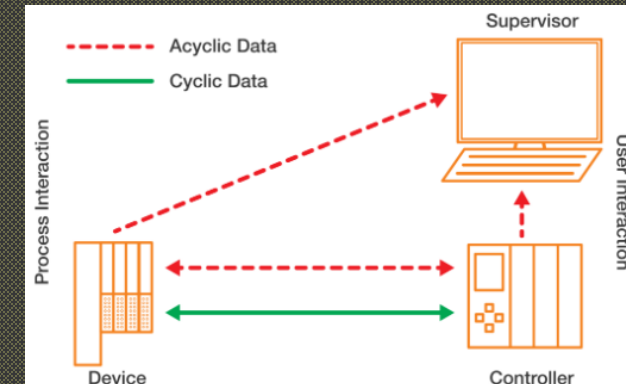
Example: A packaging machine updates motor positions every 1 ms to ensure synchronization.

3. Acyclic Communication

- Used for non-time-critical tasks (configuration, diagnostics)
- Initiated on demand
- Can take multiple cycles to complete



https://www.felser.ch/profinet-manual/pn_zyklische-daten.html



<https://toolbox.igus.com/9233/what-is-profinet>

8B.3 PROFINET Basics, Device Types, and Communication Models

PROFINET Real-Time (RT) and Isochronous Real-Time (IRT)

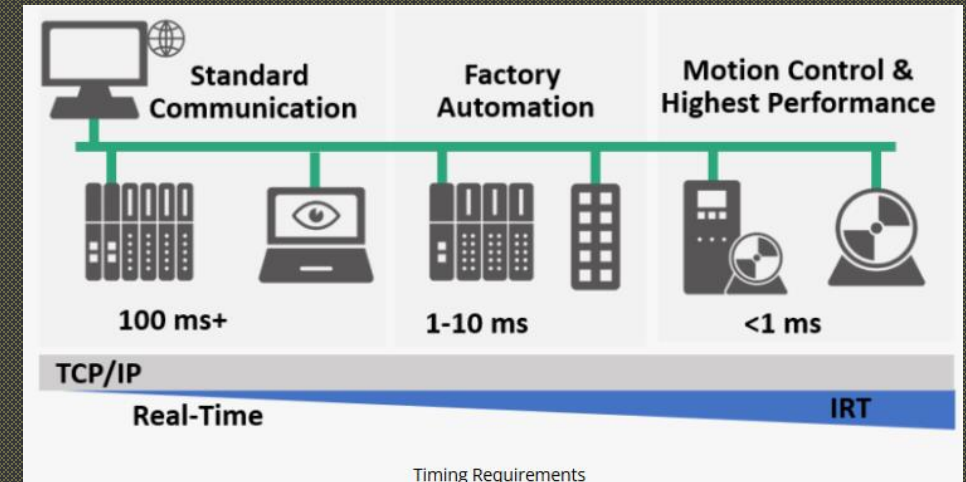
1. PROFINET RT

- For most factory automation applications
- Uses standard Ethernet switches
- Achieves cycle times from 250 μ s to 512 ms
- No special hardware needed

2. PROFINET IRT

- For motion control or multi-axis machines requiring low jitter (<1 μ s)
- Requires IRT-compatible switches and network planning
- Uses time-slot scheduling to reserve communication slots
- Bypasses TCP/IP stack for deterministic performance

Example: In a CNC machine, IRT ensures that each axis motor receives control updates at precise intervals for synchronized movement.



<https://us.profinet.com/white-papers/profinet-irt-the-solution-for-synchronous-real-time-applications/>

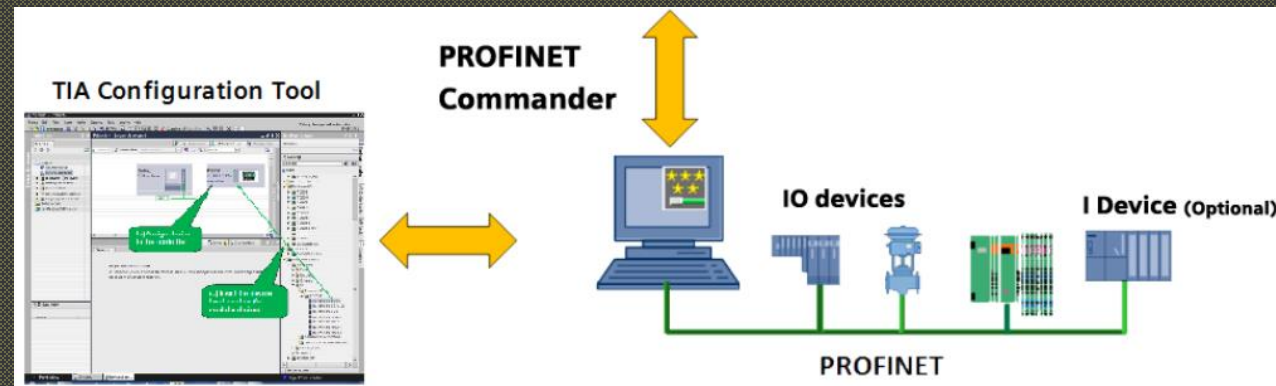
8B.3 PROFINET Basics, Device Types, and Communication Models

PROFINET Startup Process

Steps to Connect an IO-Device to an IO-Controller:

1. Discovery: IO-Controller locates devices using DCP or DHCP.
2. Connect Request: IO-Controller requests a connection.
3. Parameterization: IO-Controller checks:
 - Protocol functions supported by the device
 - Device identification (PROFINET ID)
 - Module configuration
4. Response:
 - Application Ready: Device ready for data exchange
 - Bad Connect: Device reports issues; retry sequence needed
5. Watchdog Timer: Monitors last communication; triggers reconnection if exceeded

DCP - Discovery and Configuration Protocol, used with PROFINET devices to assign network names and IP addresses on a local network.

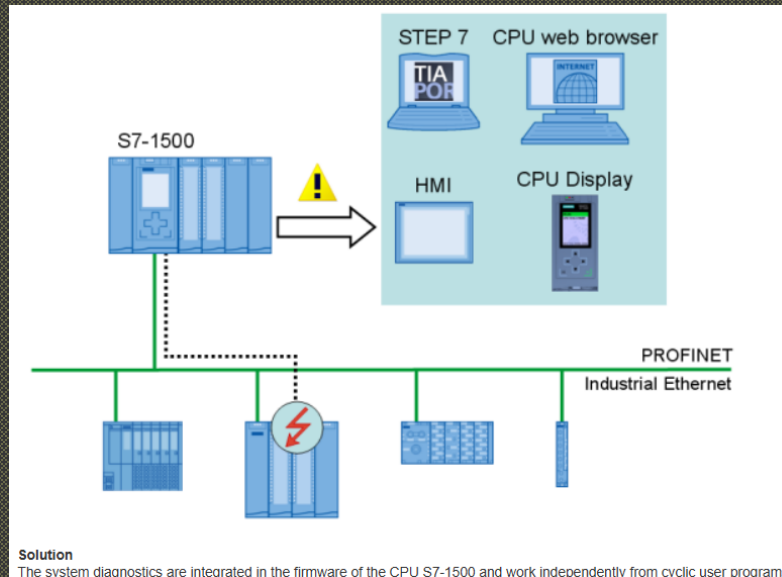


8B.3 PROFINET Basics, Device Types, and Communication Models

PROFINET Diagnostics

- Real-time channel used for sending alarms and diagnostic messages
- Prioritized messages ensure high-speed problem reporting
- Status Byte inherited from PROFIBUS shows Process Variable reliability
- Alarm Types:
 - Process Alarms: Process-related issues (e.g., overcurrent, high temperature)
 - Diagnostic Alarms: Device or network issues (e.g., wiring failure, module error)

Example: If a temperature sensor fails, the IO-Device sends a diagnostic alarm to the IO-Controller, triggering HMI alerts.



8B.3 PROFINET Basics, Device Types, and Communication Models

Addressing in PROFINET

1. Device Identification

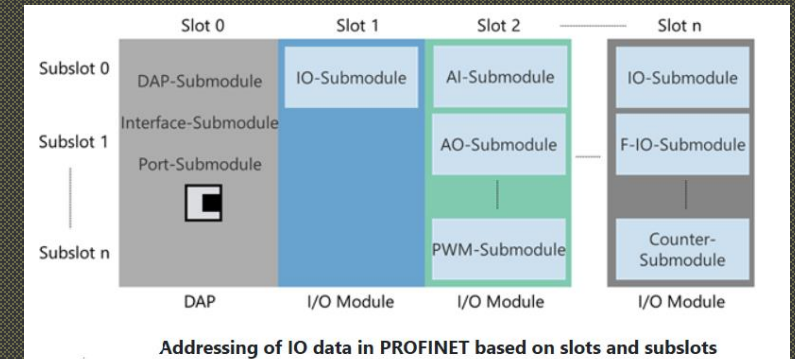
- PROFINET Name: User-friendly device identifier
- IP Address: Enables acyclic communication, network-wide access
- MAC Address: Used for fast cyclic data exchange; must be on the same subnet

2. Slot/Sub-slot/Index Model

- Slots represent modules, sub-slots represent submodules, and indexes locate specific parameters
- Facilitates cyclic and acyclic data mapping
- Simplifies integration with PROFIBUS devices

Example: A multi-channel analog input module may use:

- Slot 2 → Sub-slot 1 → Index 101 → Channel 1 voltage
- Slot 2 → Sub-slot 2 → Index 101 → Channel 2 voltage



<https://reference.opcfoundation.org/PROFINET/v101/docs/4.1>

- **DAP (Device Access Point):** The base unit of the PROFINET device. Slot 0 is always for the DAP.
- **IO-Submodule / AI-Submodule / AO-Submodule, etc.:** These are the functional blocks that represent the actual input and output data, diagnostics, and configuration for a specific channel or function.
- **Interface-Submodule & Port-Submodule:** These represent communication interfaces, like the PROFINET ports on the device, which can also be configured and diagnosed.

8B.3 PROFINET Basics, Device Types, and Communication Models

Addressing in PROFINET

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- Example: To read the current speed (analog) from MotorDrive01:
 - Controller uses: Device = MotorDrive01 (MAC 00:1A:2B:CC:DD:10)
 - Read cyclic I/O path: Slot 2 → Sub-slot 1 → Index 200 (AI Channel 1)
 - Access is fast and typically done via MAC (same subnet); IP/TCP used for non-real-time/acyclic ops.
- Example: To change a parameter (e.g., max RPM) acyclically:
 - Use IP address 192.168.10.10 (PROFINET Name or IP) and a GSDML/DTM interface
 - Access by Slot/Sub-slot and Index via engineering tool (TIA Portal)

IO-Controller (PLC)		
PROFINET Name:	PLC_Line01	
IP Address:	192.168.10.2	
MAC Address:	00:1A:2B:AA:BB:01	
Slot 0 (Device controller / base info)		
└ Sub-slot 0 (Device ID / Diagnostics)		
└ Sub-slot 1 (Status Block)		
Slot 1 (I/O Module: Digital Inputs)		
└ Sub-slot 1 (DI Channel 1..8)	Index 100	
└ Sub-slot 2 (DI Channel 9..16)	Index 101	
Slot 2 (I/O Module: Analog Inputs)		
└ Sub-slot 1 (AI Channel 1)	Index 200	
└ Sub-slot 2 (AI Channel 2)	Index 201	
Slot 3 (I/O Module: Drive Control block)		
└ Sub-slot 1 (Control Word)	Index 300	
└ Sub-slot 2 (Status Word)	Index 301	
IO-Device: MotorDrive01		
PROFINET Name:	MotorDrive01	
IP Address:	192.168.10.10	
MAC Address:	00:1A:2B:CC:DD:10	

GSDML (General Station Description Markup Language) is an XML-based file format that describes the properties of a PROFINET device.

It serves the same role that GSD files played in PROFIBUS.

A DTM (Device Type Manager) is a software component that provides a graphical and functional interface to configure, diagnose, or monitor a field device.

8B.3 PROFINET Basics, Device Types, and Communication Models

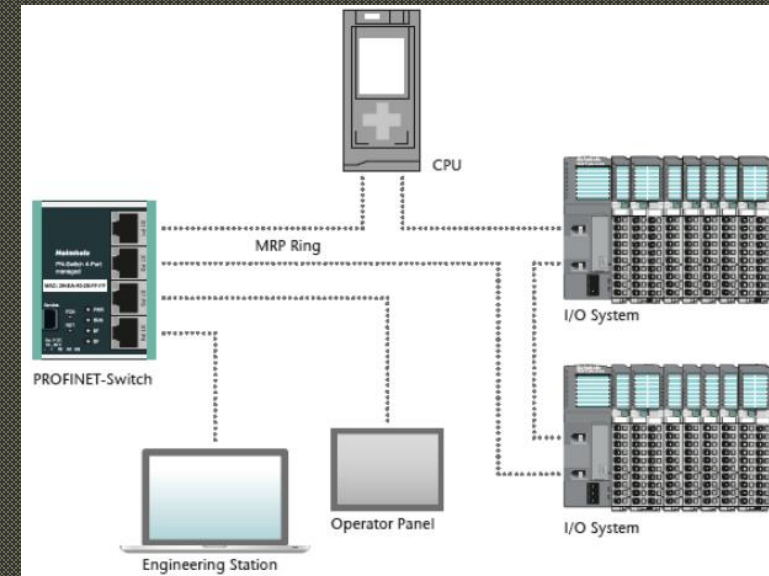
PROFINET Redundancy and Integration

- Redundant Devices: Ensures network resilience
- Star or Ring Topology: Redundant links maintain communication if a device fails
- PROFINET Proxy: Maps PROFIBUS, Modbus, or other protocols into PROFINET

Example: In a chemical plant, a ring network ensures that if one switch fails, the conveyor control continues without interruption.

Protocol: PROFINET uses the Media Redundancy Protocol (MRP).

- Roles:
 - One device is the Manager (MRM).
 - All other devices are Clients (MRC).
- Normal Operation:
 - The MRM blocks one port.
 - This turns the physical ring into a logical line, preventing data loops.
- Failure Event:
 - A cable break or device failure is detected.
 - The MRM unblocks its port.
- Redundancy:
 - Communication automatically reconverges using the alternative path in < 200ms.
 - This ensures a single point of failure does not stop the network.



8B.4 Mapping PROFINET to OSI Layers

PROFINET is built on standard Ethernet but adds real-time and industrial-specific extensions. Mapping its layers to OSI clarifies the roles of physical connections, message formats, and application functions.

PROFINET primarily uses Layers 1- 4, with specialized handling for real-time requirements.

PROFINET Physical Layer (Layer 1)

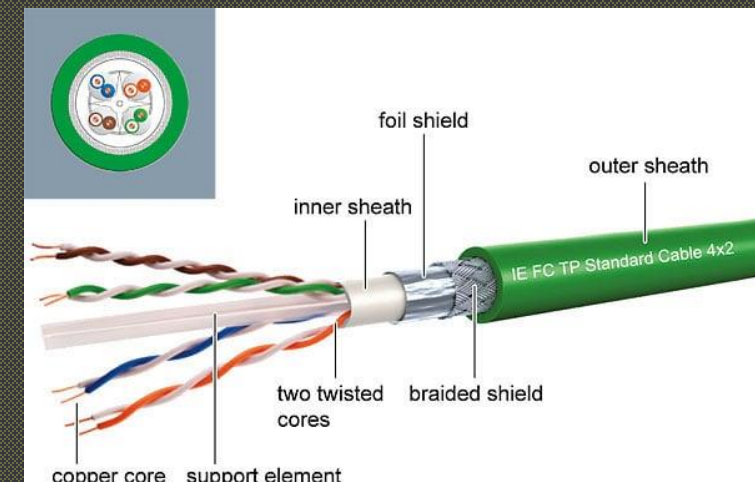
- Uses standard Ethernet cabling (twisted pair, fiber optics)
- Differential signaling for noise immunity
- Speeds:
 - 100 Mbps for PROFINET RT
 - 100/1,000 Mbps for advanced implementations
- MTL-3 and 4B/5B encoding for robust signal transmission

Key Points:

- Responsible for transmitting bits across the medium
- No awareness of message content

MLT-3 (Multi-Level Transition 3) encoding is a line code that uses three voltage levels (+V, 0, and -V) to transmit data.

4B/5B encoding is a data transmission technique that maps every 4 bits of data to a unique 5-bit code.



8B.4 Mapping PROFINET to OSI Layers

PROFINET Data Link Layer (Layer 2)

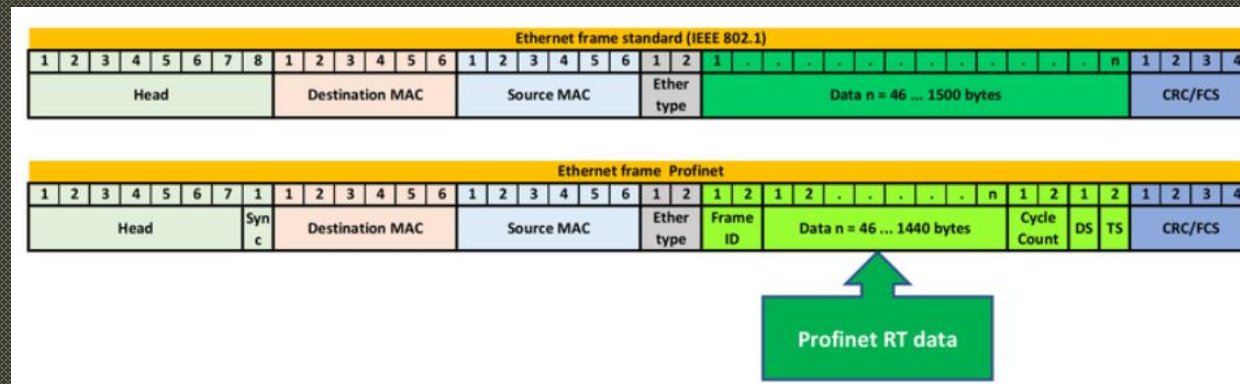
- Handles Ethernet frames, MAC addressing, and switching
- Differentiates between:
 - Real-time (RT) frames
 - Isochronous real-time (IRT) frames
 - Standard TCP/IP traffic for acyclic communication
- Manages priority tagging for RT/IRT messages (VLAN/802.1Q)
- Error detection via Frame Check Sequence (FCS)

VLANs logically split one physical network into multiple isolated ones for security and organization.

IEEE 802.1Q enables this by adding a 4-byte tag to Ethernet frames, including a 12-bit VLAN ID (up to 4094 VLANs) so switches can separate and forward traffic only within the same VLAN.

Key Points:

- Ensures deterministic communication for control loops
- Directs frames to the correct device port using MAC addresses



8B.4 Mapping PROFINET to OSI Layers

PROFINET Network Layer (Layer 3)

- Uses IP addresses for acyclic, configuration, and supervisory communication
- Supports routing between subnets (for TCP/IP traffic)
- Real-time and IRT traffic bypass Layer 3 to reduce latency

Key Points:

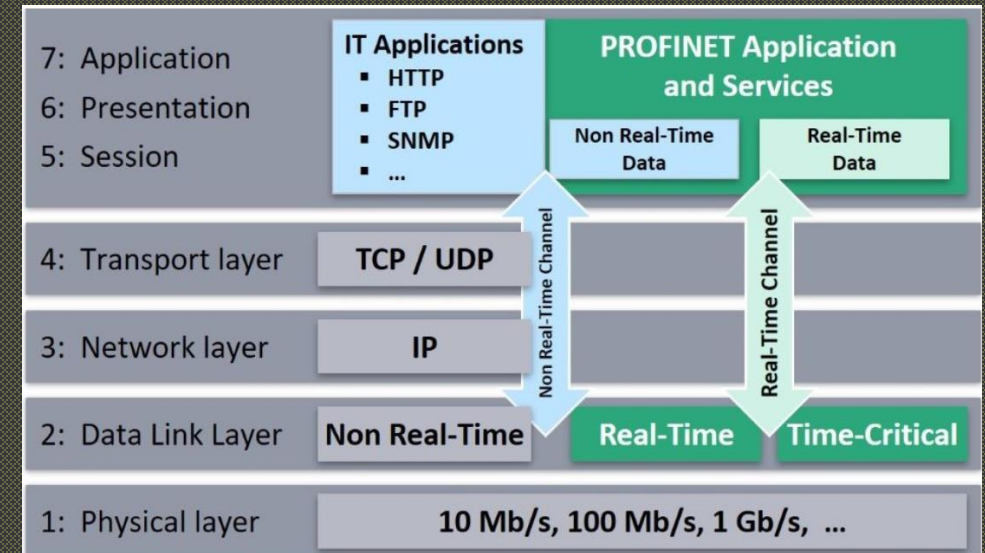
- Only used for non-time-critical traffic
- Provides interoperability with enterprise networks

PROFINET Transport Layer (Layer 4)

- TCP/UDP protocols handle non-time-critical communication
- UDP commonly used for configuration or diagnostics messages
- RT/IRT communication bypasses TCP/UDP for direct transfer to the application layer
- Ensures message integrity and minimal latency

Key Points:

- RT/IRT applications use a custom transport layer optimized for cyclic data
- TCP/UDP mainly supports IO-Supervisor to IO-Device communication



<https://www.controleng.com/using-profinet-to-improve-communication-networking/>

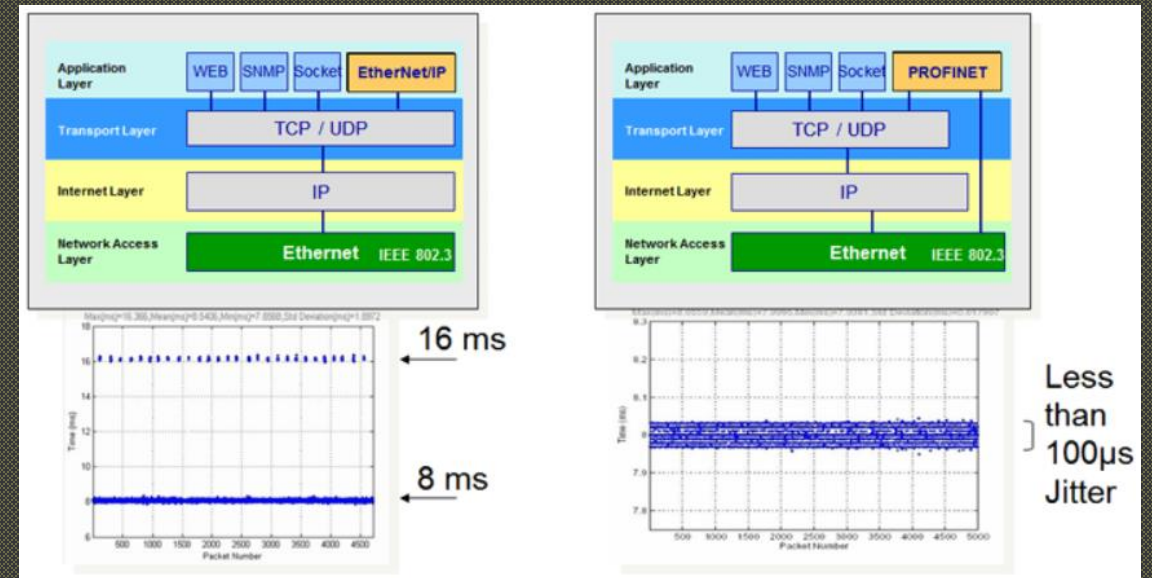
8B.4 Mapping PROFINET to OSI Layers

PROFINET Application Layer (Layer 7)

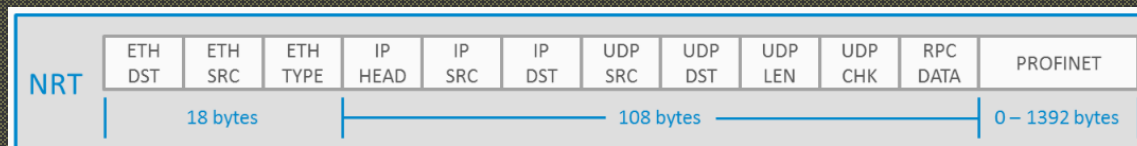
- Implements the actual industrial protocols:
 - IO-Controller ↔ IO-Device cyclic data
 - IO-Supervisor ↔ IO-Device acyclic configuration/diagnostics
- Manages:
 - Process data mapping
 - Device parameters
 - Status information
- Interfaces with Human-Machine Interfaces (HMI) or SCADA systems

Key Points:

- Provides the user-visible services of PROFINET
- Bridges engineering and field devices for monitoring and control



<https://us.profinet.com/profinet-uses-tcpip/>



PROFINET uses the NRT channel for less time-sensitive communications, like setting up a connection between a device and controller or accessing diagnostic data from an external network. But that added flexibility comes with increased data overhead – 108 extra bytes on each packet.

<https://profinetuniversity.com/profinet-basics/profinet-communication-channels/>

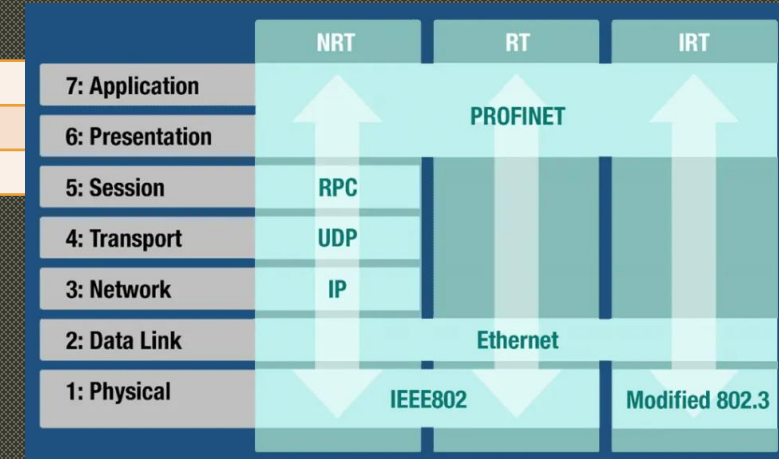
8B.4 Mapping PROFINET to OSI Layers

Real-Time and Isochronous Real-Time Traffic

Traffic Type	OSI Layers Used	Description
RT (Real-Time)	L1, L2	Standard Ethernet, prioritized for cyclic I/O updates
IRT (Isochronous Real-Time)	L1, L2 + specialized scheduling	Reserved time slots, low jitter, used for motion control

- RT bypasses Layer 3/4 for speed
- IRT ensures deterministic delivery for high-speed, synchronized operations

RPC - Remote Procedure Call - is a protocol that allows a program to execute a function on a remote computer as if it were a local one



<https://iebmedia.com/technology/industrial-ethernet/profinet-over-industrial-wlan-infrastructure/>

PROFINET is built on standard IEEE 802.3 Ethernet, but industrial automation requires much more stringent timing than typical office networks.

To achieve deterministic (predictable and repeatable) communication, PROFINET introduces two key operational modes:

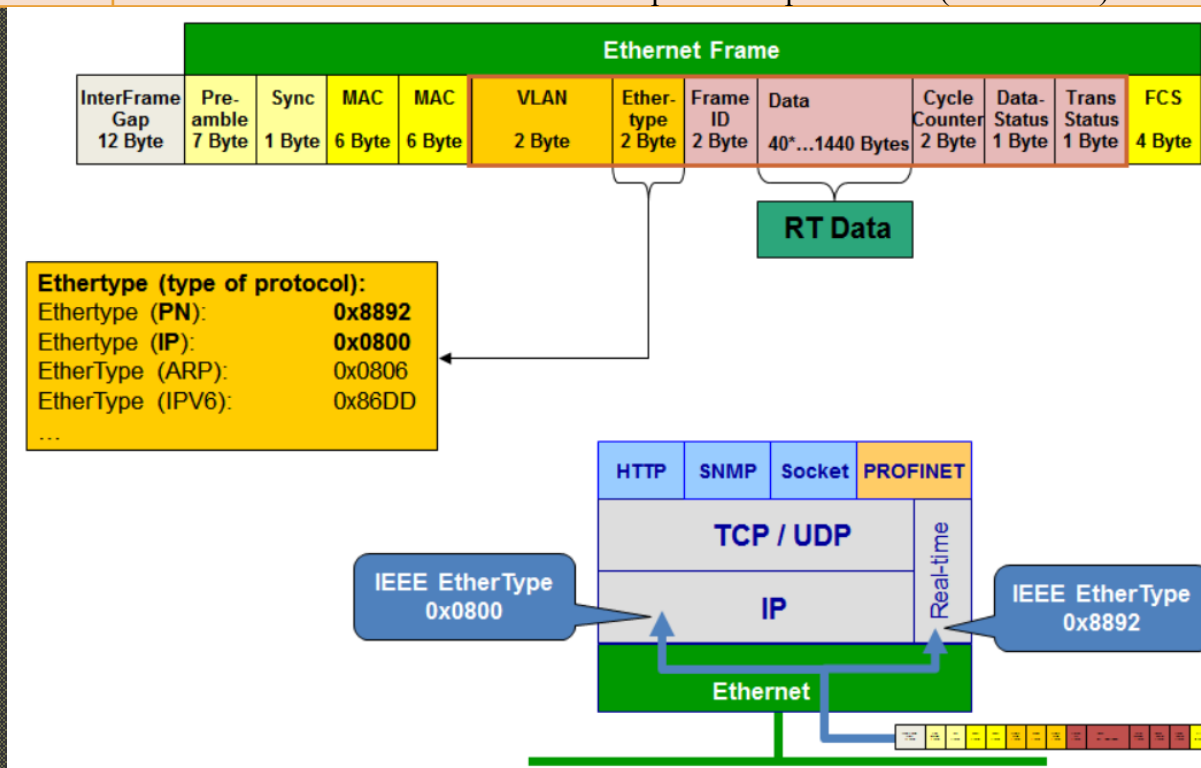
- PROFINET RT (Real-Time) – for fast and cyclic communication of process data
- PROFINET IRT (Isochronous Real-Time) – for ultra-fast, synchronized communication, mainly used in motion control

Both modes still use the Ethernet physical and data link layers (OSI Layers 1 and 2), but they bypass the higher layers of the OSI model to reduce latency and jitter.

8B.4 Mapping PROFINET to OSI Layers

So, RT and IRT frames are sent directly over Ethernet, without TCP/IP encapsulation, to achieve deterministic behavior.

OSI Layer	Function in PROFINET RT / IRT
7 – Application	Handles IO communication, parameterization, and device management.
4–6 – Transport / Session / Presentation	Bypassed for RT/IRT cyclic data (to reduce delay). Used only for acyclic communication (TCP/UDP).
3 – Network	Skipped for cyclic RT/IRT data. Used for IP-based engineering communication.
2 – Data Link	Custom PROFINET frame (EtherType 0x8892) used for deterministic transmission.
1 – Physical	Standard 100 Mbps or 1 Gbps Ethernet (IEEE 802.3).



8B.4 Mapping PROFINET to OSI Layers

PROFINET Real-Time (RT) Mode

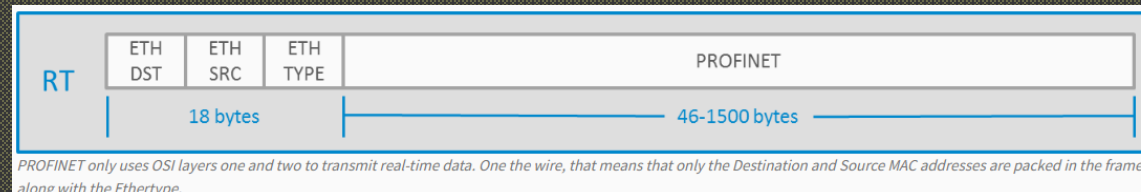
- RT mode provides deterministic cyclic communication between the controller and the field devices. It ensures predictable data exchange intervals, typically in the range of 1 to 10 milliseconds.

Implementation in the Protocol Stack

- Application Layer (Layer 7) → PROFINET IO (Cyclic Data Exchange)
 - Transport/Network Layers → Skipped (No TCP/UDP or IP)
 - Data Link Layer (Layer 2) → Ethernet Frame with Ethertype 0x8892
 - Physical Layer (Layer 1) → 100 Mbps Full Duplex Ethernet
- Key Characteristics:**
- Cycle time: ~1–10 ms
 - Jitter (timing variation): < 100 µs
 - Use case: General automation (I/O, sensors, actuators)
 - Switch type: Standard managed Ethernet switch with QoS support
 - OSI involvement: Mainly Layers 1 and 2

How It Works:

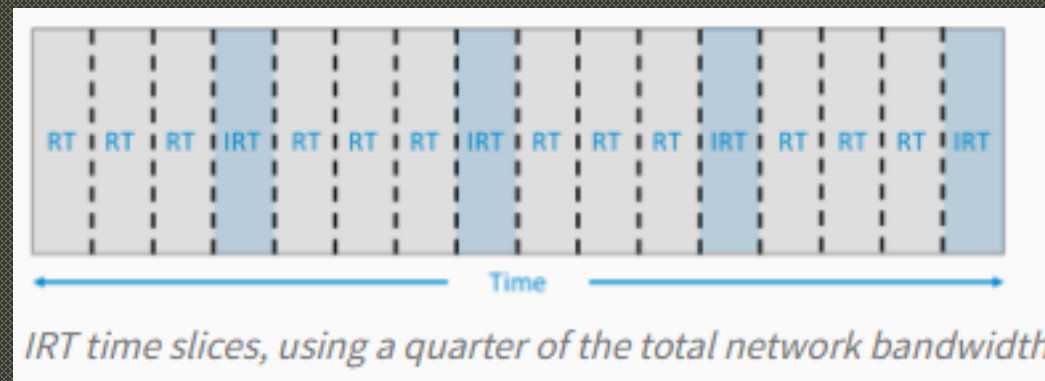
- The IO-Controller (usually a PLC) sends cyclic process data to the IO-Devices.
- Frames are transmitted using a special Ethernet Type (0x8892) reserved for PROFINET RT.
- Ethernet switches prioritize these frames using VLAN tagging (IEEE 802.1Q) and Priority Code Point (PCP) values.
- Non-real-time traffic (TCP/IP) shares the same cable but is given lower priority.
- The communication remains deterministic, as each device knows when to expect its frame.



8B.4 Mapping PROFINET to OSI Layers

PROFINET Isochronous Real-Time (IRT) Mode

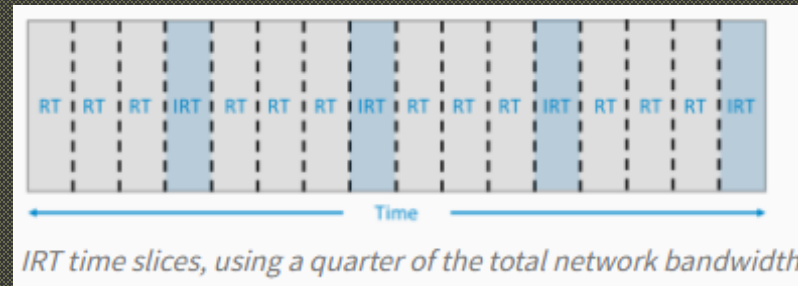
- IRT mode is designed for high-performance and synchronized motion control, where multiple axes or drives must move in perfect coordination.
It guarantees isochronous (equal-time) cycles across all network devices with microsecond-level synchronization.
- Implementation in the Protocol Stack
- Application Layer (Layer 7) → PROFINET IO (Isochronous Cyclic Data)
- Transport/Network Layers → Bypassed entirely
- Data Link Layer (Layer 2) → Enhanced Ethernet Layer with Time Scheduling
- Physical Layer (Layer 1) → 100 Mbps or 1 Gbps Ethernet (IRT-capable PHY)



8B.4 Mapping PROFINET to OSI Layers

PROFINET Isochronous Real-Time (IRT) Mode

- Bandwidth Allocation for IRT: Divide network into 4 time-slices to reserve 25% bandwidth for Isochronous Real-Time (IRT) traffic (1 slice for IRT, 3 for regular RT).
- Operation:
 - 75% of time: Normal Ethernet mode.
 - 25% of time: Buffer non-IRT traffic; only IRT frames pass during its slice.
 - After IRT slice: Release buffered traffic and resume normal operations.
- Scalability: IRT slice sized to fit exact needs, maximizing remaining bandwidth for regular traffic.
- Implementation Requirements per Switch:
 - Shared Clock: IEEE 1588 (Precision Time Protocol, PTP); PROFINET extends it via PTCP (Precision Time Control Protocol) for nanosecond-accurate delay compensation across switches and cabling, ensures synchronized entry/exit of IRT slices.
 - Buffering Circuitry: Holds non-IRT traffic arriving during IRT slice; meets PI standards and integrates into standard Ethernet switches (though advanced).



8B.4 Mapping PROFINET to OSI Layers

- PROFINET Isochronous Real-Time (IRT) Mode

How It Works:

- The IRT-capable switches and devices are synchronized using IEEE 1588 Precision Time Protocol (PTP).
 - This ensures all devices share the same clock within $\pm 1 \mu\text{s}$ accuracy.
- The Ethernet transmission cycle is divided into time slots (TDMA – Time Division Multiple Access):
 - Green phase: Reserved for IRT deterministic traffic (synchronized motion data).
 - Yellow phase: For RT frames (less critical cyclic data).
 - Blue phase: For standard TCP/IP traffic (non-time-critical).
- The Network Configuration Tool in the engineering software defines these time slots automatically based on network topology and required cycle time.
- The IO-Controller sends and receives process data precisely within the green time window.

Key Characteristics:

- Cycle time: as low as $31.25 \mu\text{s}$ (typically $250 \mu\text{s}$ to 1 ms)
- Jitter: $< 1 \mu\text{s}$
- Synchronization: via IEEE 1588 PTP
- Use case: Motion control, robotics, CNCs, synchronized drives
- Switch type: IRT-capable switch (hardware-based scheduler)
- OSI involvement: Only Layers 1 and 2

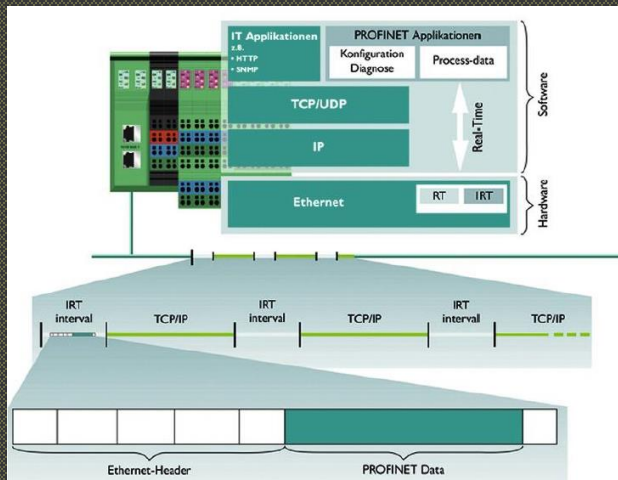
8B.4 Mapping PROFINET to OSI Layers

How RT and IRT Coexist

- Both RT and IRT traffic can exist on the same network:
 - RT traffic is handled using priority queuing.
 - IRT traffic uses reserved time slots.
 - Standard TCP/IP traffic (for engineering tools, diagnostics, etc.) runs in the remaining bandwidth.
- This layered coexistence allows a single PROFINET infrastructure to serve:
 - Real-time control loops (RT)
 - Precision motion systems (IRT)
 - Configuration and monitoring (TCP/IP)

Imagine a multi-lane highway:

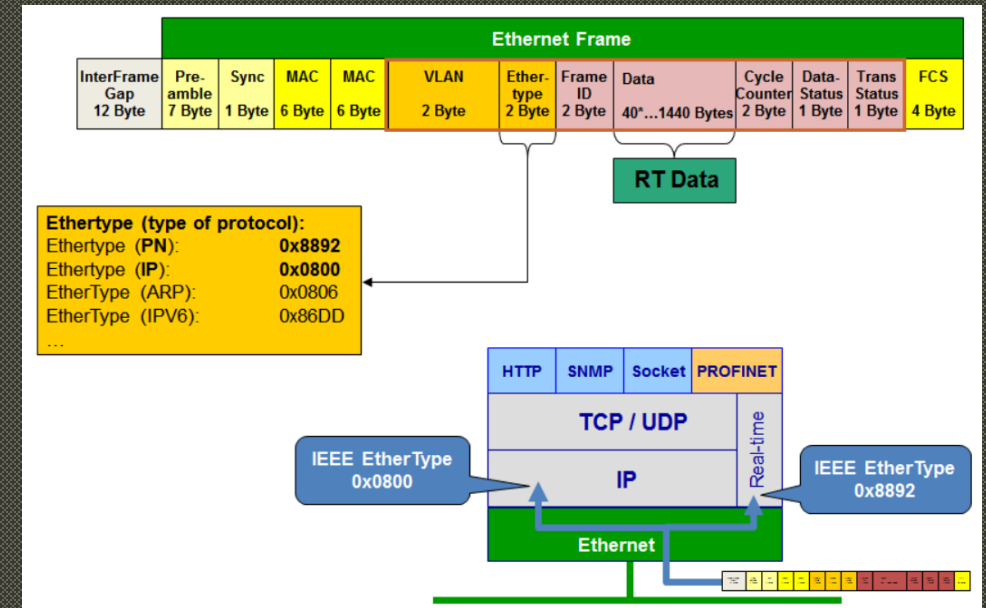
- RT Mode:**
The network gives priority lanes to PROFINET cars (control data), so they always pass through intersections faster, even during congestion.
- IRT Mode:**
The highway operates with traffic lights perfectly synchronized so that all PROFINET cars move in exact intervals — each car leaves its lane precisely on time, with no variation.
- TCP/IP Traffic:**
Regular vehicles use the remaining lanes, moving when bandwidth is available, but without guaranteed timing



Feature	PROFINET RT	PROFINET IRT
Purpose	Fast cyclic I/O	Synchronized motion control
Cycle Time	1–10 ms	31.25 μ s – 1 ms
Jitter	< 100 μ s	< 1 μ s
Synchronization	None	IEEE 1588 (PTP)
Switch Requirement	Standard managed Ethernet	Hardware IRT-capable
Traffic Handling	Priority-based	Time-slot based (TDMA)
OSI Layers Used	Layers 1–2	Layers 1–2 (enhanced)
Compatibility	All PROFINET devices	Only IRT-certified devices

8B.4 Mapping PROFINET to OSI Layers

1. Think of an Ethernet frame as a sealed envelope that contains a message. The envelope has a standard set of fields (from left to right):
2. InterFrame Gap (12 bytes)
 - The “breathing space” between two Ethernet frames. It’s like leaving a gap between envelopes so the network can process one before the next arrives.
3. Preamble (7 bytes) + Sync / SFD (1 byte)
 - These are synchronization bytes — they tell the receiver “a frame is coming” and help align clocks. Analogy: the envelope flap that signals “open now.”
4. Destination MAC (6 bytes) and Source MAC (6 bytes)
 - Physical hardware addresses of the receiver and sender (like street addresses on the envelope). Switches use these to forward the frame.
5. VLAN (2 bytes) (*optional, highlighted in orange*)
 - A tag that can carry priority info (priority code point, VLAN id). Analogy: a colored sticker on the envelope that marks it as urgent.



<https://profinetuniversity.com/industrial-automation-ethernet/network-reference-model/>

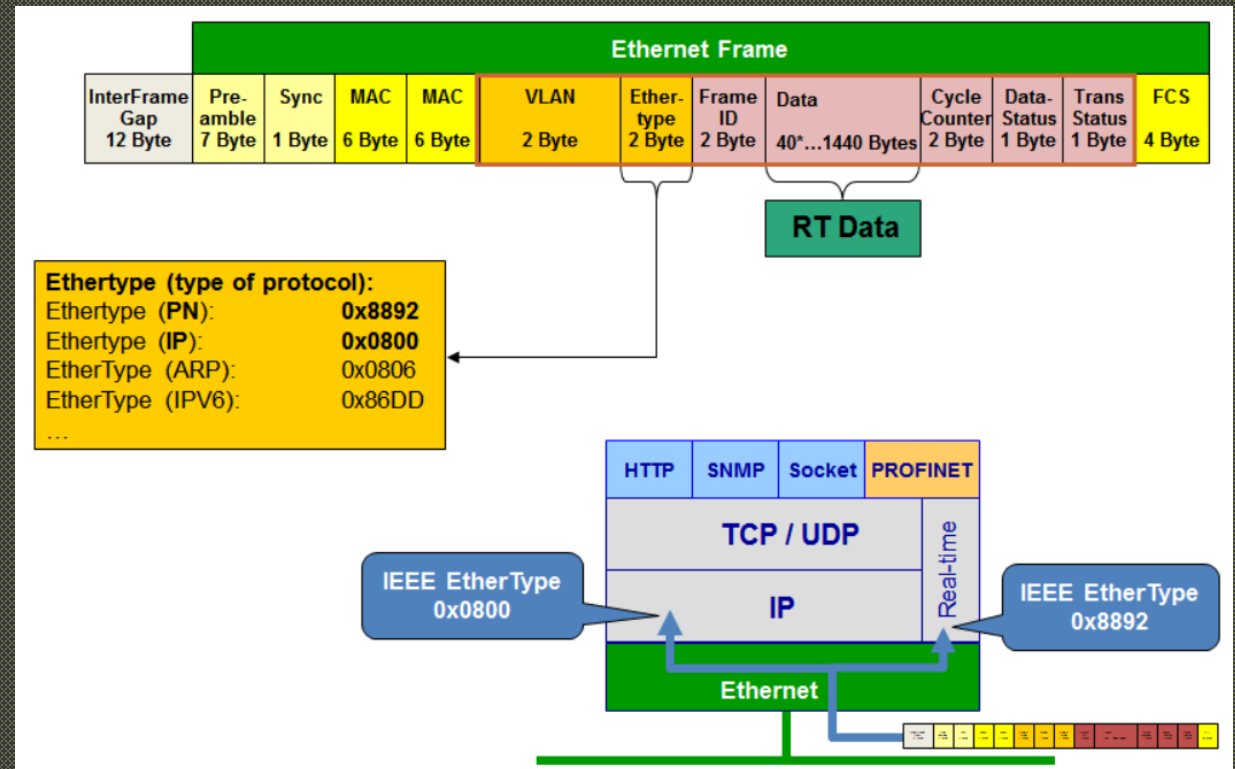
8B.4 Mapping PROFINET to OSI Layers

6. Ethertype (2 bytes)

- Tells the receiver what kind of payload is inside the frame:
 - 0x0800 → IP (normal TCP/UDP/IP traffic)
 - 0x8892 → PROFINET real-time frames (RT)
- This is the crucial decision flag: “Do we hand this envelope to the IP post office, or do we handle it as PROFINET RT?”

7. Frame ID / Data / RT Data (variable length, shown in pink)

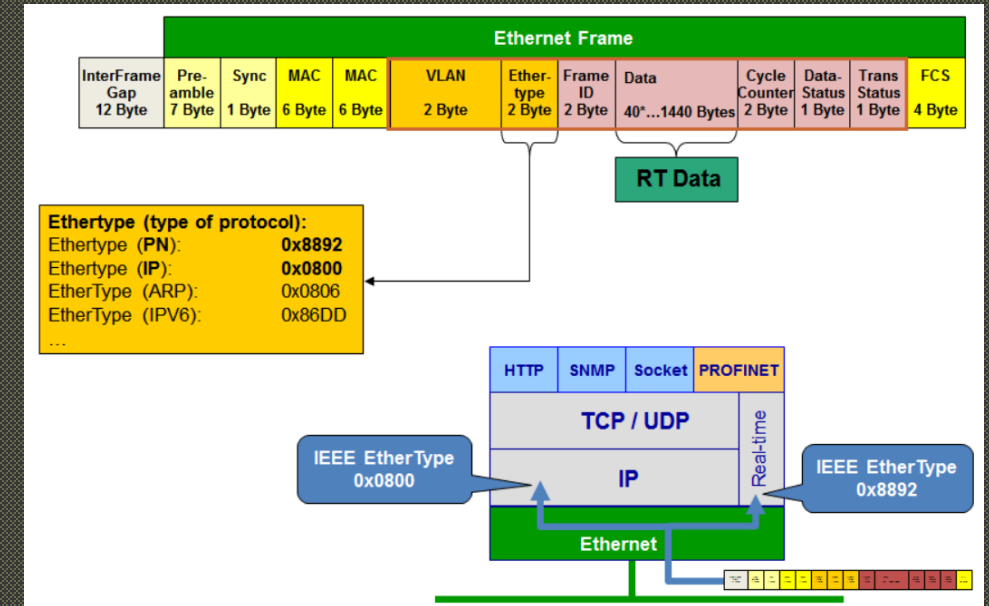
- The actual payload, the message contents. For PROFINET RT the payload includes the cyclic data and some small control fields (see next).



<https://profinetuniversity.com/industrial-automation-ethernet/network-reference-model/>

8B.4 Mapping PROFINET to OSI Layers

8. Cycle counter (2 bytes), Data-status (1 byte), Trans-status (1 byte)
 - These are PROFINET-specific fields appended in RT frames:
 - Cycle counter = a small heartbeat number that increments each I/O cycle so devices know which cycle the data belongs to (helps detect missed or out-of-order frames).
 - Data-status = indicates the quality/validity of the process value (OK, uncertain, bad).
 - Trans-status = transport/status info (e.g., whether the frame was fragmented or final).
 - Analogy: a timestamp + condition flag inside the envelope telling the receiver the freshness and health of the data.
9. FCS (Frame Check Sequence, 4 bytes)
 - A checksum so the receiver can detect if the frame contents were corrupted in transit. Like a seal that verifies nobody damaged the envelope.



<https://profinetuniversity.com/industrial-automation-ethernet/network-reference-model/>

8B.4 Mapping PROFINET to OSI Layers

Two different paths inside the protocol stack:

1. IP / TCP / UDP path (normal path)

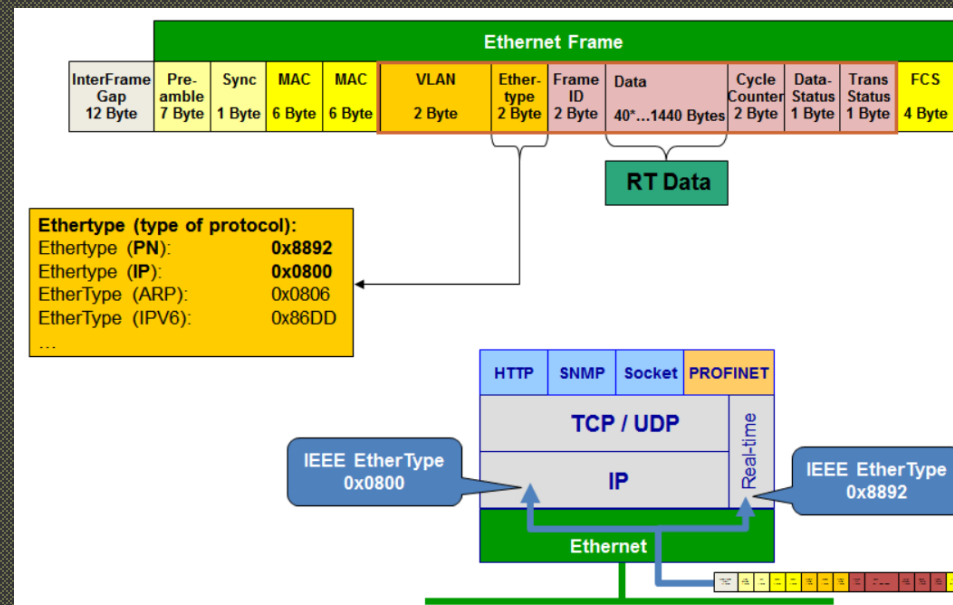
- If Ethertype = 0x0800 (IP), the frame is delivered to network Layer-3 and then to the Transport layer (TCP/UDP). This is how engineering tools, web interfaces, file transfer, and diagnostics talk to devices.
- This path is non-time-critical and can tolerate variable delay (latency).

2. PROFINET Real-time path (fast path)

- If Ethertype = 0x8892, the frame is recognized as PROFINET RT and is forwarded directly from the Data Link layer (Layer-2) to the PROFINET handling logic — bypassing IP/TCP.
- Bypassing upper layers removes processing overhead and keeps latency and jitter low.

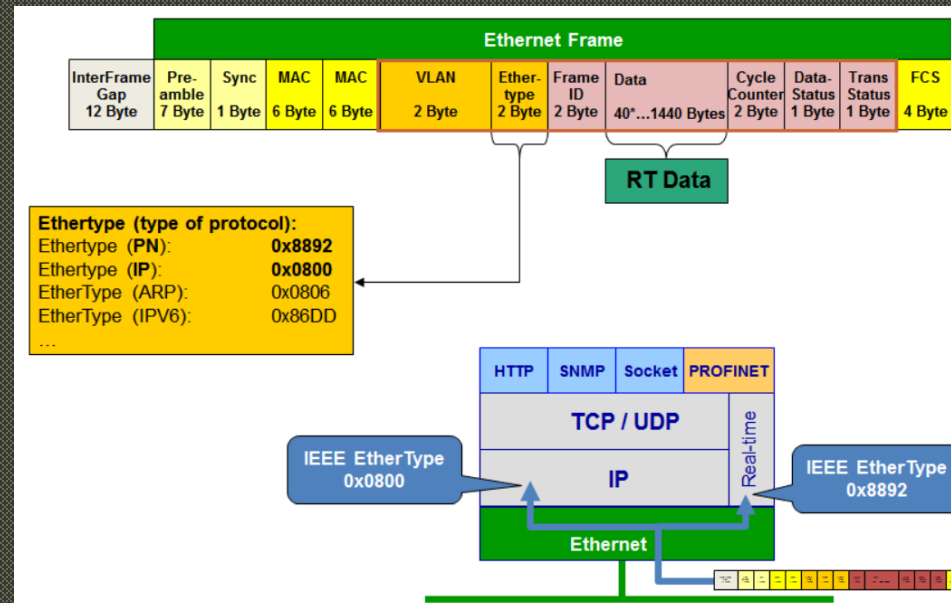
Ethertype values for different protocols:

- 0x8892 → PROFINET RT (real-time frames)
- 0x0800 → IPv4 (regular IP data)
- others follow for ARP, IPv6, etc.



8B.4 Mapping PROFINET to OSI Layers

- Ethernet carries many protocol types. The Ethertype identifies which “post office” should handle the payload.
 - Using 0x8892 for PROFINET RT tells switches and NICs “this is industrial control data, treat it as real time.”
 - Using 0x0800 for IP traffic keeps engineering/configuration packets compatible with standard network routing and tools.
 - IP/TCP add headers and processing (routing, segmentation, retransmission). That introduces unpredictable delays.
 - Real-time control needs predictable delivery with minimal jitter; skipping IP/TCP reduces CPU processing time.
 - Therefore, RT frames are kept as compact Ethernet payloads with cycle counters and status fields - fast to create and fast to parse.
-
- **MAC forwarding:** Switches forward frames based on MAC addresses quickly (no IP routing required).
 - **VLAN/PCP priority:** The VLAN tag can carry priority bits (PCP). Switches put high-priority frames into expedited queues.
 - **Switch QoS & scheduling:** Managed switches honor priority and can be configured to prioritize Ethertype 0x8892 and VLAN PCP levels.
 - **In IRT networks,** special IRT switches additionally reserve time slices (TDMA) to guarantee exclusive transmission windows - more on that if you want IRT timing details.



8B.5 PROFINET Physical Layers, Switches, and Installation Guidelines

Physical Layer Options

1. Twisted-Pair Copper (100BaseT)

- Standard Ethernet cabling (CAT5e or CAT6)
- Speeds: 100 Mbps full-duplex (standard), 10/100/1000 Mbps possible
- Maximum segment distance: 100 m between switch and device
- Uses MLT-3 and 4B/5B encoding for signal robustness

Example: Connecting PLCs, IO-devices, and HMIs inside a manufacturing cabinet using CAT5e cables.

2. Fiber Optics

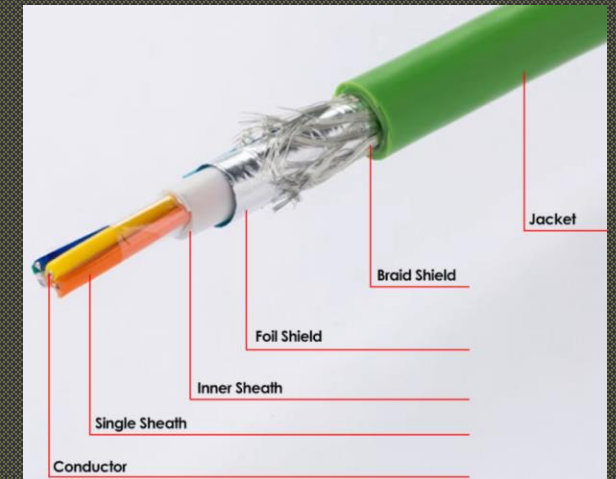
- Transmits data using light; immune to EMI and electrical noise
- Provides electrical isolation to prevent ground loops
- Distance: up to 100 km depending on quality
- Ideal for high-speed links across large plants or between buildings

Example: Linking two production halls 500 m apart where electrical noise is high.

3. Wireless Communication

- Radio frequency (Wi-Fi, proprietary industrial protocols) or infrared
- Provides cable-free communication in hard-to-wire or mobile environments
- Suitable for mobile robots, AGVs, or temporary setups
- Electrical isolation and reduced installation cost

Example: Wireless connection of a mobile conveyor system across a plant floor.



<https://www.shineindustrygroup.com/what-is-profinet-cable/>

8B.5 PROFINET Physical Layers, Switches, and Installation Guidelines

PROFINET Encoding Methods

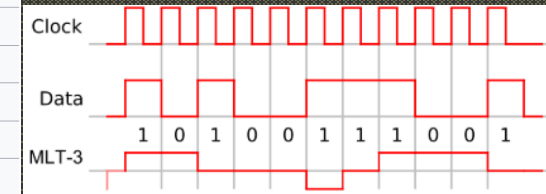
1. 4B/5B Encoding

- Converts 4-bit data blocks into 5-bit codes
- Ensures frequent transitions for synchronization and reduces DC offset
- Example: Hex “0E” (00001110) → Encoded as “11110 11100”

2. MLT-3 Encoding

- Uses three voltage levels (0, +1, -1) to transmit data
- “1” triggers a voltage transition; “0” maintains the current level
- Combined with 4B/5B to achieve 100 Mbps nominal

Data		4B5B code	Data		4B5B code
(Hex)	(Binary)		(Hex)	(Binary)	
0	0000	11110	8	1000	10010
1	0001	01001	9	1001	10011
2	0010	10100	A	1010	10110
3	0011	10101	B	1011	10111
4	0100	01010	C	1100	11010
5	0101	01011	D	1101	11011
6	0110	01110	E	1110	11100
7	0111	01111	F	1111	11101



<https://en.wikipedia.org/wiki/4B5B>

MLT-3 cycles sequentially through the voltage levels -1, 0, +1, 0. It moves to the next state to transmit a 1 bit and stays in the same state to transmit a 0 bit.

Ethernet Switches in PROFINET

- Physically connect devices
- Direct messages to correct ports
- Prioritize high-priority PROFINET messages
- Buffer lower-priority traffic

Type	Features	Usage
Unmanaged	Simple LED indicators, no advanced diagnostics	Small systems, low cost
Managed	SNMP, web access, diagnostic features	Medium to large plants
IRT-compatible	Supports Isochronous Real-Time communication	Motion control, multi-axis drives

Example: Using a managed switch in a packaging line allows network monitoring and fault diagnostics.

8B.5 PROFINET Physical Layers, Switches, and Installation Guidelines

PROFINET Network Design Rules

1. Topology Selection

- Star: Simple, switch-based
- Line: Cost-effective, uses built-in two-port switches in IO-devices
- Ring: Supports redundancy; ideal for critical networks
- Hybrid: Combination for complex layouts

2. Cable Lengths

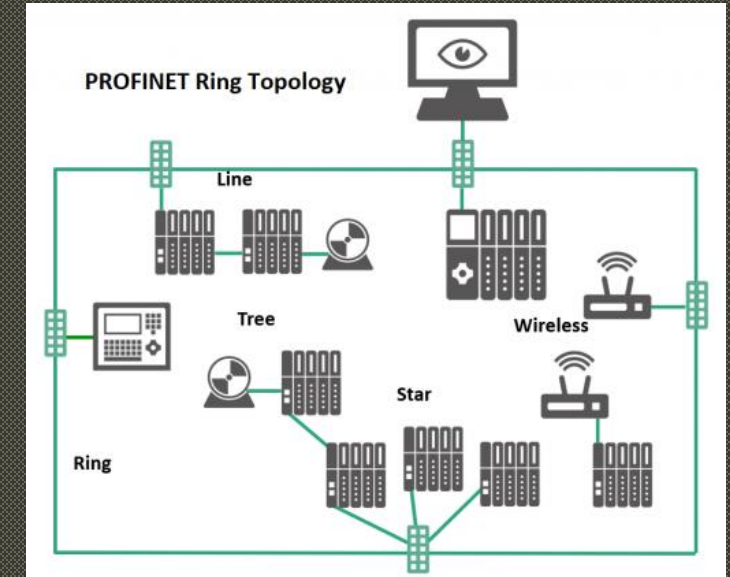
- Max 100 m between device and switch
- Longer distances require additional switches or fiber optic links

3. Switch Selection Considerations

- Full-duplex support
- Industrial-grade durability
- Port count and type
- IRT/RT support if required
- IP rating, temperature, voltage range, diagnostics, redundancy
- Standardize configuration to simplify maintenance and spare management

4. Grounding and Shielding

- Connect cable shields to ground at each device
- Use equipotential grounding for consistent plant grounding
- Reduces noise and improves signal quality



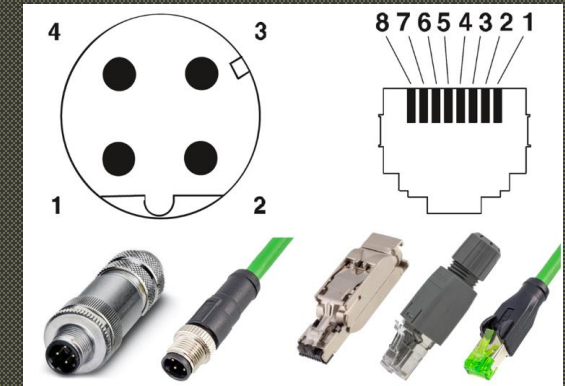
<https://us.profinet.com/profinet-media-redundancy/>

8B.5 PROFINET Physical Layers, Switches, and Installation Guidelines

IP Rating and Connector Types

IP Rating	Environment	Connector Type
IP20	Inside cabinets	Standard RJ45
IP65	Harsh, outside cabinets	Push-pull RJ45
IP67	Field devices, exposed environments	M12 circular connector

Example: Using IP67 M12 connectors for sensors mounted on factory floors with frequent water or dust exposure.



<https://www.shineindustrygroup.com/what-is-profinet-cable/>

PROFINET Redundancy

- Star Topology with Redundant Links: High availability for critical processes
- Ring Topology with Redundant Links: Data continues to flow if any one device or switch fails
- PROFINET Proxy: Integrates other protocols such as PROFIBUS, Modbus, or DeviceNet

Example: In a bottling plant, a ring network allows the conveyor and filling machine controllers to continue operating even if one switch fails.

8B.6 Network Configuration and Device Integration

Efficient configuration and integration of devices in industrial networks is critical for reliable operation. Both PROFIBUS and PROFINET use standardized files and tools to define device capabilities, communication parameters, and diagnostics.

GSD Files (General Station Description)

Overview

- Text/XML files describing device capabilities
- Define cyclic data and basic protocol information
- Used primarily by Class 1 masters (PLCs/DCS) for configuration

Key Components

- Device identity: vendor, model, and revision
- Communication properties: baud rate, address
- Module descriptions: input/output size, types, and layout
- Supported features: redundancy, diagnostics, device profiles

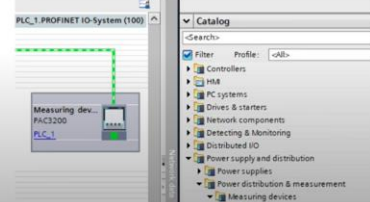
Example: Configuring a temperature transmitter on a PROFIBUS DP network:

- Load the device's GSD file in the PLC configuration software
- Assign network address and slot positions
- Map process variables (temperature) for cyclic exchange

Components Of Profinet

4. GSD Files (General Station Description)

Contain data such device characteristics, configuration and communication parameters



✓ Help reduce setup time

✓ Minimize human errors

GSD

<https://www.youtube.com/watch?v=j2Wj3SvSEdk>

8B.6 Network Configuration and Device Integration

EDD Files (Electronic Device Description)

Overview

- Describe both cyclic and acyclic capabilities of devices
- Provide detailed access for advanced configuration and diagnostics
- Primarily used by Class 2 masters (engineering workstations)

Features

- Parameter list with ranges and default values
- Diagnostic messages and alarms
- Device functions such as calibration, scaling, or mode selection

Example: Setting a pressure transmitter's measurement range using EDD in SIMATIC PDM.

PDM - Process Device Manager, a software tool used for configuring, commissioning, and maintaining intelligent field devices.

DTM (Device Type Manager)

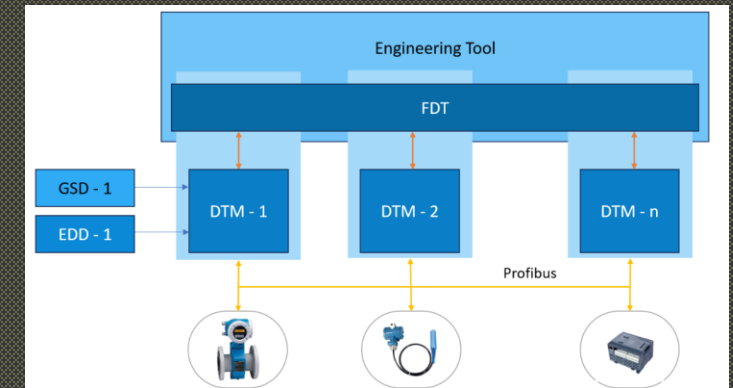
Overview

- Software component that interprets GSD or EDD files
- Provides configuration, monitoring, and diagnostic interface in Field Device Tool (FDT)
- Enables consistent user interface across multiple devices and vendors

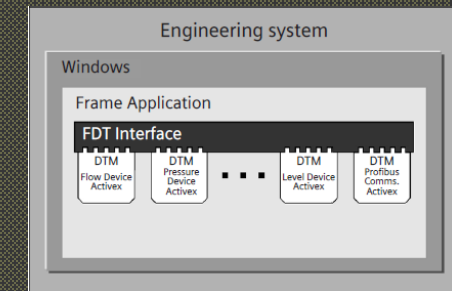
Benefits

- Simplifies configuration for mixed-vendor networks
- Standardizes maintenance procedures
- Supports both PROFIBUS and PROFINET devices

Example: Using a DTM to configure an IO-Device's cyclic data mapping for a PROFINET network.



<https://www.embien.com/industrial-insights/a-guide-to-profibus-device-integration-and-tools>



Field Device Tool (FDT)

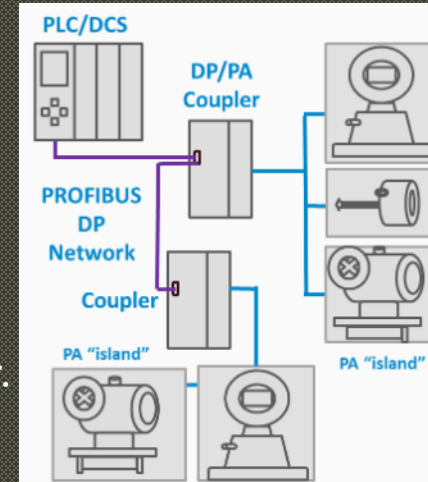
8B.6 Network Configuration and Device Integration

Device Integration

1. PROFIBUS DP/PA Integration

- DP-V0/V1 defines the basic communication structure
- PA devices use MBP (Manchester Bus Powered) or RS-485 for physical layer
- Device profiles ensure standardization across manufacturers
- Process variable mapping: Analog (4–20 mA) and digital (ON/OFF) values

Example: Integrating a flow transmitter (PA device) into a PROFIBUS DP segment using a DP/PA coupler.

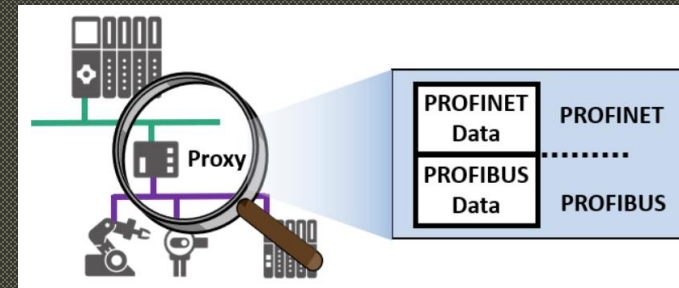


<https://profinetuniversity.com/process-automation/pa-process-automation-basics/>

2. PROFINET Device Integration

- Uses GSDML files (XML format) for devices
- IO-Controllers (PLCs/DCS) assign IP addresses and cyclic data slots
- Slot/sub-slot/index addressing maps process variables
- Proxy gateways enable PROFIBUS devices to appear as PROFINET devices

Example: Connecting a PROFIBUS temperature transmitter to a PROFINET controller via a proxy gateway.



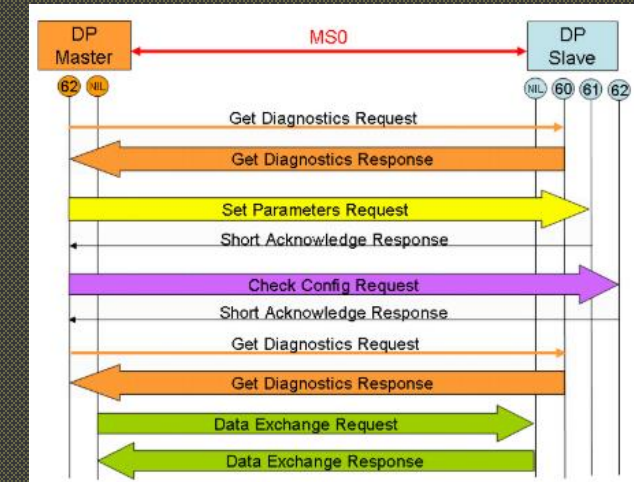
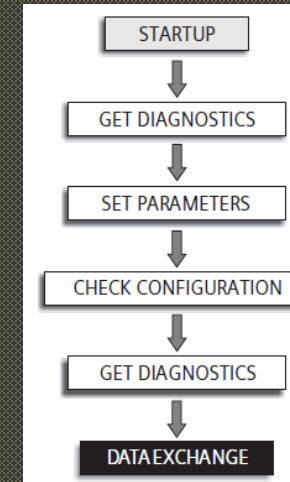
<https://us.profinet.com/technology/profinet/>

8B.6 Network Configuration and Device Integration

Start-Up and Commissioning Sequence

1. PROFIBUS Start-Up

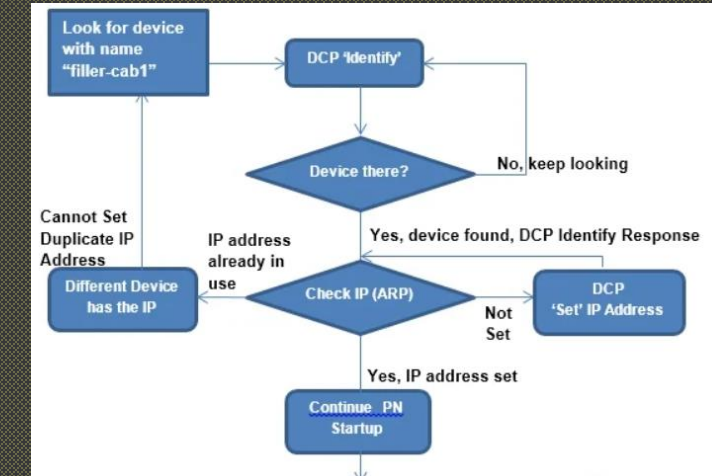
- Master scans the network for slaves
- Checks diagnostics for existing master assignment
- Sets communication parameters and assigns addresses
- Configures device modules and confirms readiness
- Enters cyclic data exchange
- Watchdog timer ensures ongoing communication



<https://www.felser.ch/profibus-manual/initialisierungsssequenz.html>

2. PROFINET Start-Up

- Discovery via DCP or DHCP to locate IO-Device
- Connect request sent from IO-Controller
- Parameterization of supported functions, identification, and module configuration
- Device sends Application Ready response
- Cyclic data exchange begins
- Watchdog timers ensure continuous communication



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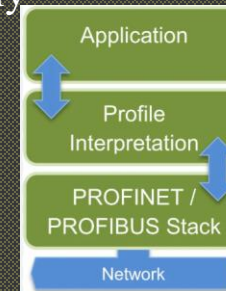
<https://profinetuniversity.com/naming-addressing/profinet-dcp/>

8B.6 Network Configuration and Device Integration

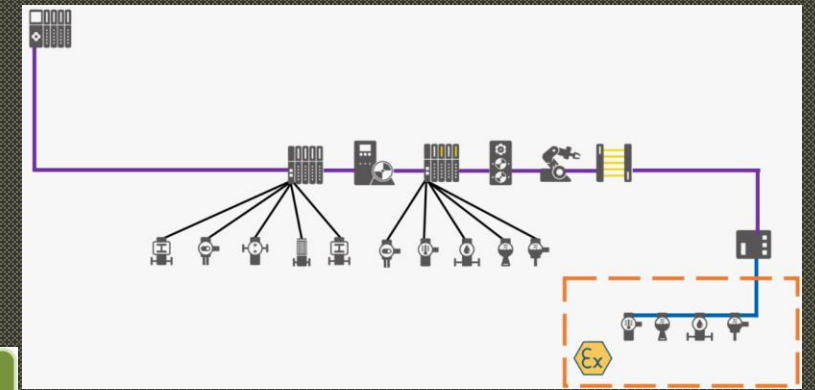
Device Profiles and Standardization

- Profiles define:
 - Data structure and format
 - Standard parameters and functions
 - Health and status indicators
- Simplifies device replacement and multi-vendor integration
- Reduces engineering time, programming complexity, and errors

Example: Switching a level transmitter from Vendor A to Vendor B requires only changing the GSD/GSDML file and keeping addresses consistent.



<https://profinetuniversity.com/application-profiles/application-profiles-explained/>



<https://us.profinet.com/profibus-device-types-explained-master-vs-devices/>

Diagnostics and Maintenance

1. PROFIBUS Diagnostics

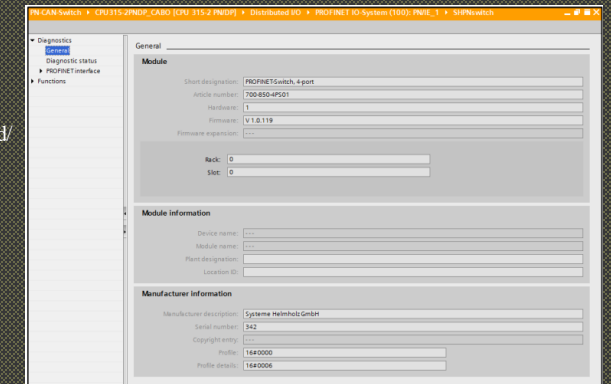
- Cyclic and acyclic diagnostic messages
- Status byte and extended status flags
- Alerts for process alarms, hardware errors, and device health

Example: Master detects sensor failure (Loss of Echo) and triggers maintenance alert.

2. PROFINET Diagnostics

- Real-time diagnostic messages for process and hardware issues
- Alarm types: Process alarms vs. Diagnostic alarms
- Status byte indicates process variable reliability

Example: IO-Device detects over-current in a motor drive and immediately notifies IO-Controller.



https://www.helmholz.de/media/pdf/ea/da/03/Advantages_of_a_PROFINET-Switch_-_Version_4_EN.pdf

8B.6 Network Configuration and Device Integration

Real-Time Communication (RT)

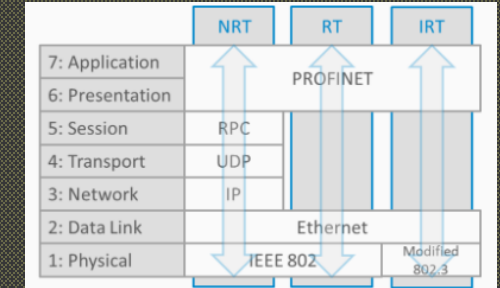
Overview

- Standard Ethernet is not deterministic; RT ensures time-critical data arrives predictably.
- Uses a dedicated real-time channel bypassing TCP/IP for time-sensitive I/O.

Key Features

- Cycle times: 250 μ s – 512 ms (typical)
- High-priority Ethernet frames for cyclic I/O
- Standard switches can handle RT for most factory applications
- Watchdog timers monitor communication intervals

Example: Reading digital sensor inputs and controlling actuators in a packaging line every 1 ms using RT communication.



<https://profinetuniversity.com/profinet-basics/profinet-communication-channels/>

Isochronous Real-Time (IRT)

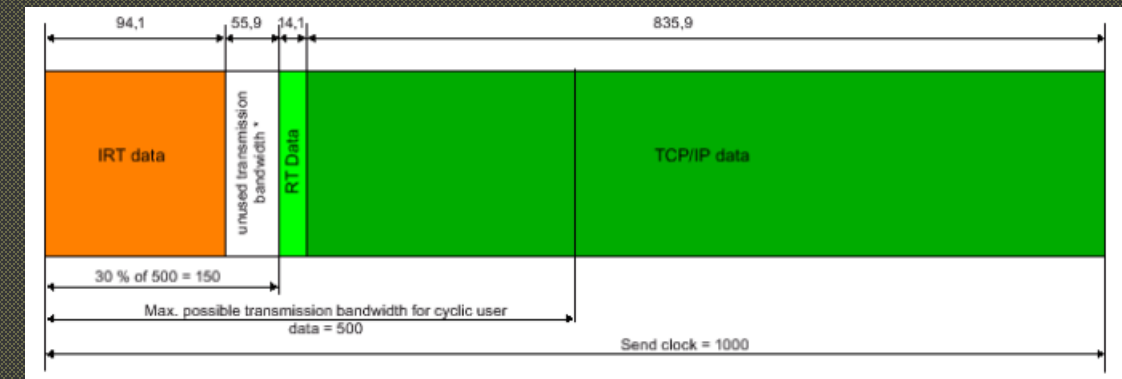
Overview

- IRT extends RT for high-speed, multi-axis motion control
- Guarantees deterministic delivery with low jitter (<1 μ s)
- Uses time-slot scheduling to reserve bandwidth for IRT traffic

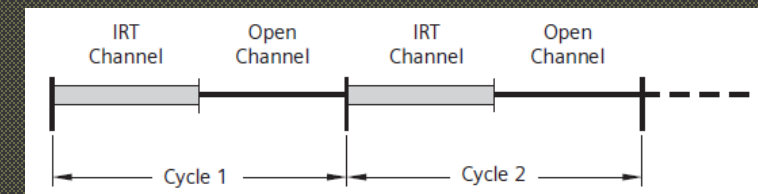
Operation

- Each device receives a time slot for communication
- Synchronizes clocks across all devices
- Standard Ethernet channels handle non-critical traffic concurrently

Example: Coordinating a 6-axis robotic arm in a production cell with <1 μ s jitter.



<https://support.industry.siemens.com/cs/mdm/19292127?c=36667789707&lc=en-AF>



8B.6 Network Configuration and Device Integration

Time-Sensitive Networking (TSN)

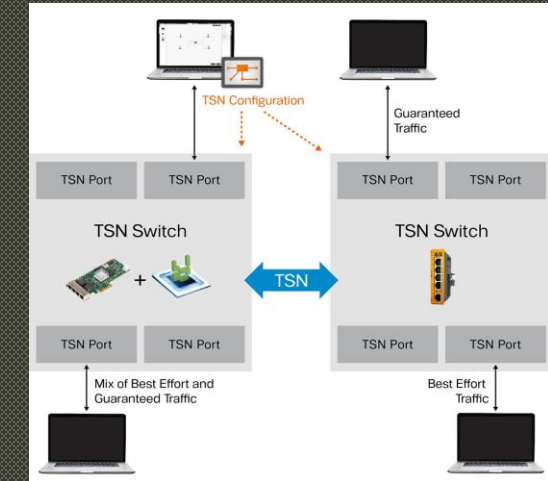
Overview

- TSN enhances PROFINET RT and IRT with standard IEEE 802.1 features
- Prioritizes traffic, reduces latency, and ensures deterministic timing
- Useful for large, mixed traffic networks

Key Features

- Time synchronization across devices
- Traffic shaping to avoid congestion
- Quality of Service (QoS) for critical messages

Example: Integrating high-speed vision sensors with robotic actuators on the same network without timing conflicts.

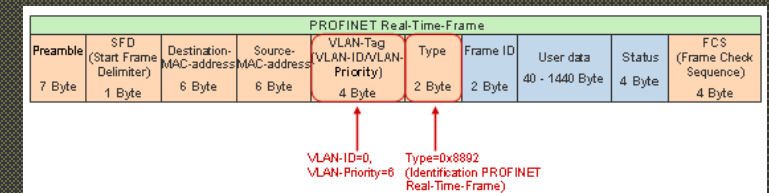


<https://iebmmedia.com/technology/tsn/converged-it-ot-networks-using-time-sensitive-networking-tsn-technology/>

PROFINET Prioritization

- Real-time and IRT messages have the highest priority
- Non-time-critical traffic is delayed until real-time frames are delivered
- Switches use frame tagging to identify priority levels

Example: Alarm messages are assigned lower priority than cyclic motor control frames.



8B.7 The Three PROFINET Communication Channels

PROFINET traffic can be divided into three distinct “paths” based on timing requirements and frame type:

Communication Type	Abbreviation	Purpose	Timing Requirement	Typical Cycle Time
Non Real-Time	NRT	Configuration, parameterization, diagnostics	Non-deterministic	100 ms – seconds
Real-Time	RT	Cyclic process data (sensors, actuators)	Deterministic	1–10 ms
Isochronous Real-Time	IRT	High-precision motion control, synchronized drives	Deterministic and synchronized	31.25 μ s – 1 ms

All three channels use Ethernet (IEEE 802.3) as the base, but each mode has different handling inside the protocol stack and different Ethertypes / scheduling mechanisms.

Communication Flow by Type

1. Non Real-Time (NRT)

- Used for acyclic and non-time-critical tasks: configuration, diagnostics, and file transfers.
- Uses TCP/IP and UDP/IP protocols (Ethernet = 0x0800).
- Passes through the full OSI stack.
- Can traverse routers and firewalls.
- Delays are variable (non-deterministic).

Example:

- Engineering PC uses Siemens TIA Portal or another configuration tool to download a PLC program.
- The PC communicates with the PLC over standard IP (Ethernet 0x0800) and TCP port 34962.
- Timing is not critical, milliseconds or even seconds of delay are acceptable.

8B.7 The Three PROFINET Communication Channels

Communication Flow by Type

2. Real-Time (RT)

- Used for cyclic data exchange between controller and field devices (sensors, valves, PLC I/O modules).
- Ethertype = 0x8892 (PROFINET frame).
- Bypasses TCP/UDP/IP layers, handled directly at Ethernet Layer 2.
- Switches prioritize it using VLAN priority bits (PCP = 6 or 7).
- Each RT frame includes Cycle Counter, Data Status, and Transfer Status bytes.

PCP - Priority Code Point, a 3-bit field within an Ethernet frame's VLAN (Virtual Local Area Network) tag (IEEE 802.1Q) used for prioritizing network traffic.

Example:

- A PLC sends output data to a valve manifold every 2 ms and receives input feedback (pressure, position).
- The communication is cyclic and deterministic, each device knows exactly when its frame will arrive.
- VLAN tagging ensures these frames are transmitted before normal IP traffic.

3. Isochronous Real-Time (IRT)

- For ultra-precise, synchronized control, such as multi-axis motion control or robotics.
- Still Ethertype = 0x8892, but handled by an IRT-capable Layer 2 stack with hardware time scheduling.
- The network cycle is divided into time slots (TDMA):
 - Green phase: IRT traffic (highest priority)
 - Yellow phase: RT traffic (medium priority)
 - Blue phase: TCP/IP or NRT traffic (lowest priority)
- Synchronization across devices achieved using IEEE 1588 Precision Time Protocol (PTP).
- Cycle times can be as low as 31.25 μ s with jitter < 1 μ s.

TDMA – Time Division Multiple Access

Example:

- A motion control PLC coordinates three servo drives so they rotate perfectly in phase (e.g., in a printing press or robotic arm).
- All drives and controller clocks are synchronized to within 1 μ s.
- The IRT scheduler ensures each frame is sent at an exact time: no collisions, no jitter.

8B.7 The Three PROFINET Communication Channels

How PROFINET Tells Them Apart

1. At the Frame Level: PROFINET distinguishes between these communication types by looking at specific fields inside the Ethernet frame header and the payload:

Field	Used By	Description
Ethertype	RT vs NRT	Identifies protocol type. 0x8892 = PROFINET RT/IRT; 0x0800 = standard IP/TCP (NRT).
Frame ID	RT vs IRT	Inside the PROFINET frame (after Ethertype), defines the frame function and priority.
VLAN Priority (PCP bits)	RT	Used to give RT frames higher priority than IP frames in switches (IEEE 802.1Q).
Scheduling / Time Slot Assignment	IRT	Determines when IRT frames can be transmitted, based on TDMA scheduling.

So, the Ethertype tells if the frame is PROFINET or IP-based, and the Frame ID + VLAN priority + timing window determine if it is RT or IRT.

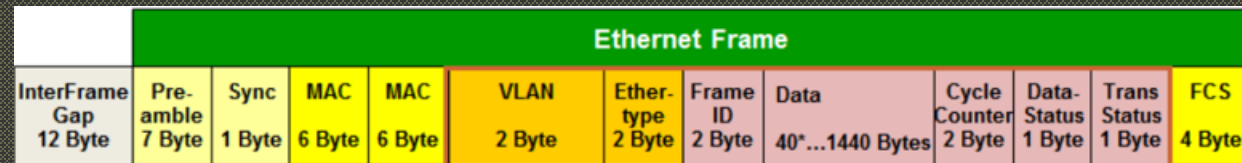
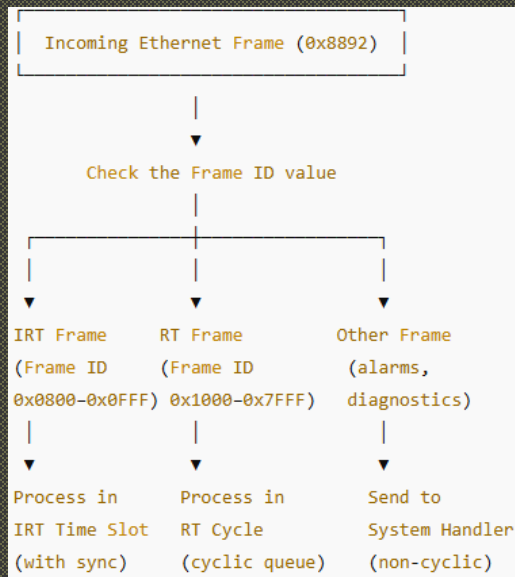
2. Inside the Device Stack: Every PROFINET device (IO-Device, IO-Controller, etc.) has a communication stack that can separate and process incoming traffic based on these indicators:
 - If the frame's Ethertype = 0x0800, it goes to the IP stack for NRT traffic.
 - If the frame's Ethertype = 0x8892, it goes to the PROFINET real-time (Layer 2) handler.
 - Within the PROFINET handler, the Frame ID determines whether it is a regular RT frame or an IRT-scheduled frame.This separation ensures that real-time cyclic data bypasses slow IP routing and non-real-time data (like engineering₃ configuration) doesn't block control traffic.

8B.7 The Three PROFINET Communication Channels

All of these can travel through the same Ethernet cable, but switches and devices know which queue or time slot to use:

- VLAN priority or IRT time slot ensures proper timing.
- PROFINET protocol stack separates frames based on Ethertype and Frame ID.
- CPU and firmware schedule tasks according to priority.

Network Segment	Example Traffic	Handling Method	Timing Priority
Engineering station to PLC	Downloading a project	NRT (IP/TCP)	Low
PLC to I/O modules	Cyclic process data	RT (Ethertype 0x8892)	Medium
PLC to Servo drives	Motion commands	IRT (scheduled TDMA slot)	Highest



Feature	NRT	RT	IRT
Ethertype	0x0800 (IP)	0x8892	0x8892
Protocol Layers Used	Full TCP/IP	Layer 2 only	Layer 2 + TDMA scheduling
Synchronization	None	None	IEEE 1588 PTP
Cycle Time	100 ms – seconds	1–10 ms	31.25 µs – 1 ms
Jitter	Unbounded	<100 µs	<1 µs
Typical Use	Engineering, HMI	I/O data	Motion control
Switch Type	Standard Ethernet	Managed Ethernet	IRT-capable switch

REFERENCES

- Catching the Process Fieldbus - An Introduction to PROFIBUS and PROFINET -2nd edition
James Powell, Henry Vandelinde

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Thank you!

Discussions?